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Lab 01

INVERTED PENDULUM

Davide Burlotti, davide.burlotti@mail.polimi.it

Matteo Antonio Castiglioni, matteoantonio.castiglioni@mail.polimi.it

Ginevra Ceolato, ginevra.ceolato@mail.polimi.it

Mattia de Guarinoni, mattia.deguarinoni@mail.polimi.it

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1. Nonlinear modeling

In this section, using $x(t)$ and $\theta(t)$ as generalized coordinates, the nonlinear model of the cart-pendulum system subjected to the control action $i(t)$ and disturbance $d(t)$ is derived. The electrodynamics of the motor is neglected and the actuator is assumed to generate a horizontal force on the cart proportional to the current, i.e. $f(t) = \alpha_m \cdot i(t)$, with $\alpha_m = k_t/r$.

1.1. Implicit form

The nonlinear equations of motion in implicit form are derived using *Lagrange's* equations.

By performing a kinematic analysis, expressions for the position, velocity, and acceleration were derived for both the cart

$$\begin{cases} \vec{x}_{cart} = x \cdot \underline{i} \\ \vec{v}_{cart} = \dot{x} \cdot \underline{i} \\ \vec{a}_{cart} = \ddot{x} \cdot \underline{i} \end{cases} \quad (1)$$

and a generic point p located along the rod

$$\begin{cases} \vec{x}_p = [x - s \cdot \sin(\theta)] \cdot \underline{i} + [s \cdot \cos(\theta) + H] \cdot \underline{j} \\ \vec{v}_p = [\dot{x} - s \cdot \dot{\theta} \cdot \cos(\theta)] \cdot \underline{i} - [s \cdot \dot{\theta} \cdot \sin(\theta)] \cdot \underline{j} \\ \vec{a}_p = [\ddot{x} - s \cdot \ddot{\theta} \cdot \cos(\theta) + s \cdot \dot{\theta}^2 \cdot \sin(\theta)] \cdot \underline{i} - [s \cdot \ddot{\theta} \cdot \sin(\theta) + s \cdot \dot{\theta}^2 \cdot \cos(\theta)] \cdot \underline{j} \end{cases} \quad (2)$$

The quantities needed to establish the Lagrangian of the system were subsequently determined (T, V, D, Q_θ, Q_x).

To simplify the notation, the following quantities were introduced:

$$I_0 = \int_0^l \rho(s) \cdot ds, \quad I_1 = \int_0^l \rho(s) \cdot s \cdot ds, \quad I_2 = \int_0^l \rho(s) \cdot s^2 \cdot ds, \quad \alpha_0 = \int_0^l d_0(s) \cdot ds, \quad \alpha_1 = \int_0^l d_0(s) \cdot s \cdot ds$$

Kinetic Energy

$$\begin{aligned} T &= \frac{1}{2} \cdot M \cdot \dot{x}^2 + \frac{1}{2} \int_0^l \rho(s) \cdot ds \cdot \vec{v}_p^2 \\ &= \frac{1}{2} \cdot M \cdot \dot{x}^2 + \frac{1}{2} \int_0^l \rho(s) \cdot ds \cdot \dot{x}^2 + \frac{1}{2} \int_0^l \rho(s) \cdot s^2 \cdot ds \cdot \dot{\theta}^2 - \int_0^l \rho(s) \cdot s \cdot ds \cdot \dot{x} \cdot \dot{\theta} \cdot \cos(\theta) \\ &= \frac{1}{2} \cdot (M + I_0) \cdot \dot{x}^2 + \frac{1}{2} \cdot I_2 \cdot \dot{\theta}^2 - I_1 \cdot \dot{x} \cdot \dot{\theta} \cdot \cos(\theta) \end{aligned}$$

Potential Energy

$$\begin{aligned} V &= M \cdot g \cdot h_c + \int_0^l \rho(s) \cdot g \cdot (s \cdot \cos(\theta) + H) \cdot ds \\ &= M \cdot g \cdot h_c + g \cdot \cos(\theta) \cdot \int_0^l \rho(s) \cdot s \cdot ds + g \cdot H \cdot \int_0^l \rho(s) \cdot ds \\ &= M \cdot g \cdot h_c + I_1 \cdot g \cdot \cos(\theta) + I_0 \cdot g \cdot H \end{aligned}$$

Dissipated Energy

$$D = \frac{1}{2} \cdot c \cdot \dot{x}^2 + \frac{1}{2} \cdot b \cdot \dot{\theta}^2$$

Non-conservative forces $\rightarrow$ principle of virtual work

$$\text{Actuator: } \delta w_{act} = (\alpha_m \cdot i(t)) \cdot \delta_x$$

$$\begin{aligned} \text{Disturbance: } \delta w_{dist} &= \int_0^l [(-d_0(s) \cdot d(t)) \cdot [\delta x - s \cdot \delta_\theta \cdot \cos(\theta)]] \cdot ds \\ &= -d(t) \cdot \delta_x \cdot \int_0^l d_0 \cdot ds + d(t) \cdot \cos(\theta) \cdot \delta_\theta \cdot \int_0^l d_0 \cdot s \cdot ds \\ &= -\alpha_0 \cdot d(t) \cdot \delta_x + \alpha_1 \cdot d(t) \cdot \cos(\theta) \cdot \delta_\theta \end{aligned}$$

$$\Rightarrow \delta W = Q_\theta \cdot \delta_\theta + Q_x \cdot \delta_x = \alpha_m \cdot i(t) \cdot \delta_x - \alpha_0 \cdot d(t) \cdot \delta_x + \alpha_1 \cdot d(t) \cdot \cos(\theta) \cdot \delta_\theta$$

$$\begin{cases} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) - \frac{\partial T}{\partial \theta} + \frac{\partial V}{\partial \theta} + \frac{\partial D}{\partial \dot{\theta}} = Q_\theta \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} + \frac{\partial D}{\partial \dot{x}} = Q_x \end{cases}$$

The terms of the first equation are:

$$\begin{cases} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) = I_2 \cdot \ddot{\theta} - I_1 \cdot \ddot{x} \cdot \cos(\theta) + I_1 \cdot \dot{x} \cdot \dot{\theta} \cdot \sin(\theta) \\ \frac{\partial T}{\partial \theta} = I_1 \cdot \dot{x} \cdot \dot{\theta} \cdot \sin(\theta) \\ \frac{\partial V}{\partial \theta} = -I_1 \cdot g \cdot \sin(\theta) \\ \frac{\partial D}{\partial \dot{\theta}} = b \cdot \dot{\theta} \\ Q_\theta = \alpha_1 \cdot d(t) \cdot \cos(\theta) \end{cases}$$

$$\Rightarrow I_2 \cdot \ddot{\theta} - I_1 \cdot \ddot{x} \cdot \cos(\theta) + b \cdot \dot{\theta} - I_1 \cdot g \cdot \sin(\theta) = \alpha_1 \cdot d(t) \cdot \cos(\theta)$$

The terms of the second equation are:

$$\begin{cases} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) = (M + I_0) \cdot \ddot{x} - I_1 \cdot \ddot{\theta} \cdot \cos(\theta) + I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) \\ \frac{\partial T}{\partial x} = 0 \\ \frac{\partial V}{\partial x} = 0 \\ \frac{\partial D}{\partial \dot{x}} = c \cdot \dot{x} \\ Q_x = \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) \end{cases}$$

$$\Rightarrow (M + I_0) \cdot \ddot{x} - I_1 \cdot \ddot{\theta} \cdot \cos(\theta) + I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) + c \cdot \dot{x} = \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)$$

Finally, the resulting equations are:

$$\begin{cases} I_2 \cdot \ddot{\theta} - I_1 \cdot \ddot{x} \cdot \cos(\theta) + b \cdot \dot{\theta} - I_1 \cdot g \cdot \sin(\theta) = \alpha_1 \cdot d(t) \cdot \cos(\theta) \\ (M + I_0) \cdot \ddot{x} - I_1 \cdot \ddot{\theta} \cdot \cos(\theta) + I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) + c \cdot \dot{x} = \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) \end{cases} \quad (3)$$

1.2. Explicit form

To rearrange the equations in explicit form by isolating the accelerations $\ddot{x}(t)$ and $\ddot{\theta}(t)$ the matrix form was used:

$$\begin{bmatrix} I_2 & -I_1 \cdot \cos(\theta) \\ -I_1 \cdot \cos(\theta) & M + I_0 \end{bmatrix} \cdot \begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} I_1 \cdot g \cdot \sin(\theta) - b \cdot \dot{\theta} + \alpha_1 \cdot d(t) \cdot \cos(\theta) \\ -I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) \end{bmatrix} \quad (4)$$

For simplicity, the forcing vector of (4) is referred to as the vector $\begin{bmatrix} L_1 \\ L_2 \end{bmatrix}$

Solving the linear system yields the accelerations:

$$\begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \frac{1}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \cdot \begin{bmatrix} M + I_0 & I_1 \cdot \cos(\theta) \\ I_1 \cdot \cos(\theta) & I_2 \end{bmatrix} \cdot \begin{bmatrix} L_1 \\ L_2 \end{bmatrix}$$

$$\begin{cases} \ddot{\theta} = \frac{(M + I_0) \cdot L_1 + I_1 \cdot \cos(\theta) \cdot L_2}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ \ddot{x} = \frac{I_1 \cdot \cos(\theta) \cdot L_1 + I_2 \cdot L_2}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \end{cases} \quad (5)$$

1.3. State-space form

The system is expressed in state-space form

$$\vec{x}'(t) = f(\vec{x}(t), i(t), d(t), t) \quad (6)$$

using the following state vector:

$$\vec{x}(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} \quad (7)$$

Substituting the state variables into (5) yields:

$$\begin{aligned} \ddot{\theta} &= \frac{(M + I_0) \cdot [I_1 \cdot g \cdot \sin(\theta) - b \cdot \dot{\theta} + \alpha_1 \cdot d(t) \cdot \cos(\theta)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ &+ \frac{I_1 \cdot \cos(\theta) \cdot [-I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ &= \frac{(M + I_0) \cdot [I_1 \cdot g \cdot \sin(x_3(t)) - b \cdot x_4(t) + \alpha \cdot d(t) \cdot \cos(x_1(t))]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \\ &+ \frac{I_1 \cdot \cos(x_3(t)) \cdot [-I_1 \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \end{aligned} \quad (8)$$

and

$$\begin{aligned} \ddot{x} &= \frac{I_1 \cdot \cos(\theta) \cdot L_1 + I_2 \cdot L_2}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ &= \frac{I_1 \cdot \cos(\theta) \cdot [I_1 \cdot g \cdot \sin(\theta) - b \cdot \dot{\theta} + \alpha \cdot d(t) \cdot \cos(\theta)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ &+ \frac{I_2 \cdot [-I_1 \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(\theta)^2} \\ &= \frac{I_1 \cdot \cos(x_3(t)) \cdot [I_1 \cdot g \cdot \sin(x_3(t)) - b \cdot x_4(t) + \alpha \cdot d(t) \cdot \cos(x_3(t))]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \\ &+ \frac{I_2 \cdot [-I_1 \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \end{aligned} \quad (9)$$

The system in state-space form is expressed as

$$\begin{aligned} \vec{x}'(t) &= \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \\ \dot{x}_4(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ x_4(t) \\ \frac{I_1 \cdot \cos(x_3(t)) \cdot [I_1 \cdot g \cdot \sin(x_3(t)) - b \cdot x_4(t) + \alpha_1 \cdot d(t) \cdot \cos(x_3(t))] + I_2 \cdot [-I_1 \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \\ \frac{(M + I_0) \cdot [I_1 \cdot g \cdot \sin(x_3(t)) - b \cdot x_4(t) + \alpha \cdot d(t) \cdot \cos(x_1(t))] + I_1 \cdot \cos(x_3(t)) \cdot [-I_1 \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - \alpha_0 \cdot d(t)]}{I_2 \cdot (M + I_0) - I_1^2 \cdot \cos(x_3(t))^2} \end{bmatrix} \end{aligned}$$

2. Nonlinear Analysis and Simulation (MATLAB)

A MATLAB script was developed to simulate the nonlinear dynamics of a rod with constant linear density and constant spatial disturbance $d_0 = 0.30$ N/m. Accordingly, the kinematic model was updated and the Lagrange equations were reformulated to account for these specific conditions.

The expressions for the position, velocity, and acceleration were re-derived for both the cart

$$\begin{cases} \vec{x}_{cart} = x \cdot \underline{i} \\ \vec{v}_{cart} = \dot{x} \cdot \underline{i} \\ \vec{a}_{cart} = \ddot{x} \cdot \underline{i} \end{cases} \quad (10)$$

and the center of mass of the rod G

$$\begin{cases} \vec{x}_G = [x - \frac{l}{2} \cdot \sin(\theta)] \cdot \underline{i} + [\frac{l}{2} \cdot \cos(\theta) + H] \cdot \underline{j} \\ \vec{v}_G = [\dot{x} - \frac{l}{2} \cdot \dot{\theta} \cdot \cos(\theta)] \cdot \underline{i} - [\frac{l}{2} \cdot \dot{\theta} \cdot \sin(\theta)] \cdot \underline{j} \\ \vec{a}_G = [\ddot{x} - \frac{l}{2} \cdot \ddot{\theta} \cdot \cos(\theta) + \frac{l}{2} \cdot \dot{\theta}^2 \cdot \sin(\theta)] \cdot \underline{i} - [\frac{l}{2} \cdot \ddot{\theta} \cdot \sin(\theta) + \frac{l}{2} \cdot \dot{\theta}^2 \cdot \cos(\theta)] \cdot \underline{j} \end{cases} \quad (11)$$

The quantities needed to establish the Lagrangian of the system were subsequently re-determined (T, V, D, Q_θ, Q_x).

Kinetic Energy

$$\begin{aligned} T &= \frac{1}{2} \cdot M \cdot \dot{x}^2 + \frac{1}{2} \cdot \rho \cdot l \cdot \left(\dot{x}^2 + \frac{l^2}{4} \cdot \dot{\theta}^2 - l \cdot \dot{x} \cdot \dot{\theta} \cdot \cos \theta \right) + \frac{l}{2} \cdot I_G \cdot \dot{\theta}^2 \\ &= \frac{1}{2} \cdot M \cdot \dot{x}^2 + \frac{1}{2} \cdot m \cdot \left(\dot{x}^2 + \frac{l^2}{4} \cdot \dot{\theta}^2 - l \cdot \dot{x} \cdot \dot{\theta} \cdot \cos \theta \right) + \frac{l}{2} \cdot \frac{\rho l^3}{12} \cdot \dot{\theta}^2 \\ &= \frac{1}{2} \cdot (M + m) \cdot \dot{x}^2 + \frac{1}{2} \cdot \left(\frac{1}{3} \cdot m \cdot l^2 \right) \cdot \dot{\theta}^2 - \frac{1}{2} \cdot m \cdot l \cdot \dot{x} \cdot \dot{\theta} \cdot \cos \theta \end{aligned}$$

Potential Energy

$$\begin{aligned} V &= M \cdot g \cdot h_c + m \cdot g \cdot \left(\frac{l}{2} \cdot \cos(\theta) + H \right) \\ &= M \cdot g \cdot h_c + m \cdot g \cdot H + m \cdot g \cdot \frac{l}{2} \cdot \cos(\theta) \end{aligned}$$

Dissipated Energy

$$D = \frac{1}{2} \cdot c \cdot \dot{x}^2 + \frac{1}{2} \cdot b \cdot \dot{\theta}^2$$

Non-conservative forces $\rightarrow$ principle of virtual work

$$\text{Actuator: } \delta w_{act} = (\alpha_m \cdot i(t)) \cdot \delta x$$

$$\begin{aligned} \text{Disturbance: } \delta w_{dist} &= \int_0^l [(-d_0 \cdot d(t)) \cdot [\delta x - \frac{l}{2} \cdot \delta \theta \cdot \cos(\theta)]] \cdot ds \\ &= -d_0 \cdot d(t) \cdot l \cdot \delta x + d_0 \cdot \frac{l^2}{2} \cdot d(t) \cdot \cos(\theta) \cdot \delta \theta \\ &= \left(-d_0 \cdot d(t) \cdot \delta x + d_0 \cdot \frac{l}{2} \cdot d(t) \cdot \cos(\theta) \cdot \delta \theta \right) \cdot l \end{aligned}$$

$$\Rightarrow \delta W = Q_\theta \cdot \delta \theta + Q_x \cdot \delta x = \alpha_m \cdot i(t) \cdot \delta x + \left(-d_0 \cdot d(t) \cdot \delta x + d_0 \cdot \frac{l}{2} \cdot d(t) \cdot \cos(\theta) \cdot \delta \theta \right) \cdot l$$

Lagrange equations

$$\begin{cases} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) - \frac{\partial T}{\partial \theta} + \frac{\partial V}{\partial \theta} + \frac{\partial D}{\partial \dot{\theta}} = Q_\theta \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} + \frac{\partial D}{\partial \dot{x}} = Q_x \end{cases}$$

The terms of the first equation are:

$$\left\{ \begin{array}{l} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}} \right) = \frac{1}{3} \cdot m \cdot l^2 \cdot \ddot{\theta} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{x} \cdot \cos(\theta) + \frac{1}{2} \cdot m \cdot l \cdot \dot{x} \cdot \dot{\theta} \cdot \sin(\theta) \\ \frac{\partial T}{\partial \theta} = \frac{1}{2} \cdot m \cdot l \cdot \dot{x} \cdot \dot{\theta} \cdot \sin(\theta) \\ \frac{\partial V}{\partial \theta} = -\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) \\ \frac{\partial D}{\partial \dot{\theta}} = b \cdot \dot{\theta} \\ Q_\theta = d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta) \end{array} \right.$$

$$\Rightarrow \frac{1}{3} \cdot m \cdot l^2 \cdot \ddot{\theta} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{x} \cdot \cos(\theta) - \frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) + b \cdot \dot{\theta} = d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta)$$

The terms of the second equation are:

$$\left\{ \begin{array}{l} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) = (M + m) \cdot \ddot{x} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{\theta} \cdot \cos(\theta) + \frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) \\ \frac{\partial T}{\partial x} = 0 \\ \frac{\partial V}{\partial x} = 0 \\ \frac{\partial D}{\partial \dot{x}} = c \cdot \dot{x} \\ Q_x = \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \end{array} \right.$$

$$\Rightarrow (M + m) \cdot \ddot{x} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{\theta} \cdot \cos(\theta) + \frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) + c \cdot \dot{x} = \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t)$$

Finally, the resulting equations are:

$$\left\{ \begin{array}{l} \frac{1}{3} \cdot m \cdot l^2 \cdot \ddot{\theta} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{x} \cdot \cos(\theta) - \frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) + b \cdot \dot{\theta} = d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta) \\ (M + m) \cdot \ddot{x} - \frac{1}{2} \cdot m \cdot l \cdot \ddot{\theta} \cdot \cos(\theta) + \frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) + c \cdot \dot{x} = \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \end{array} \right. \quad (12)$$

To rearrange the equations in explicit form by isolating the accelerations $\ddot{x}(t)$ and $\ddot{\theta}(t)$ the matrix form was used:

$$\begin{bmatrix} \frac{1}{3} \cdot m \cdot l^2 & -\frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) \\ -\frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) & M + m \end{bmatrix} \cdot \begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) - b \cdot \dot{\theta} + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta) \\ -\frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \end{bmatrix} \quad (13)$$

For simplicity, the forcing vector of (13) is referred to as the vector $\begin{bmatrix} L_3 \\ L_4 \end{bmatrix}$

Solving the linear system yields the accelerations:

$$\begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \frac{1}{\frac{1}{3} \cdot m \cdot (M + m) - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \cdot \begin{bmatrix} M + m & \frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) \\ \frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) & \frac{1}{3} \cdot m \cdot l^2 \end{bmatrix} \cdot \begin{bmatrix} L_3 \\ L_4 \end{bmatrix}$$

$$\left\{ \begin{array}{l} \ddot{\theta} = \frac{(M + m) \cdot L_3 + \frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) \cdot L_4}{\frac{1}{3} \cdot m \cdot (M + m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \\ \ddot{x} = \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos \theta \cdot L_3 + \frac{1}{3} \cdot m \cdot l^2 \cdot L_4}{\frac{1}{3} \cdot m \cdot (M + m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \end{array} \right. \quad (14)$$

The system is then expressed in state-space form

$$\vec{\dot{x}}(t) = f(\vec{x}(t), i(t), d(t), t) \quad (15)$$

using the following state vector:

$$\vec{x}(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} \quad (16)$$

Substituting the state variables into (14) yields:

$$\begin{aligned}
\ddot{\theta} &= \frac{(M+m) \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) - b \cdot \dot{\theta} + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \\
&+ \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos(\theta) \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \\
&= \frac{(M+m) \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(x_3(t)) - b \cdot x_4(t) + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(x_3(t)) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2} \\
&+ \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos(x_3(t)) \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2}
\end{aligned} \tag{17}$$

and

$$\begin{aligned}
\ddot{x} &= \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos \theta \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(\theta) - b \cdot \dot{\theta} + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(\theta) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \\
&+ \frac{\frac{1}{3} \cdot m \cdot l^2 \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot \dot{\theta}^2 \cdot \sin(\theta) - c \cdot \dot{x} + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(\theta)^2} \\
&= \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos(x_3(t)) \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(x_3(t)) - b \cdot x_4(t) + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(x_3(t)) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2} \\
&+ \frac{\frac{1}{3} \cdot m \cdot l^2 \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2}
\end{aligned} \tag{18}$$

The system in state-space form is expressed as

$$\begin{aligned}
\vec{\dot{x}}(t) &= \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \\ \dot{x}_4(t) \end{bmatrix} = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \\ \dot{\theta}(t) \\ \ddot{\theta}(t) \end{bmatrix} \\
&= \begin{bmatrix} x_2(t) \\ \frac{\frac{1}{2} \cdot m \cdot l \cdot \cos(x_3(t)) \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(x_3(t)) - b \cdot x_4(t) + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(x_3(t)) \right) + \frac{1}{3} \cdot m \cdot l^2 \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2} \\ x_4(t) \\ \frac{(M+m) \cdot \left(-\frac{1}{2} \cdot m \cdot g \cdot l \cdot \sin(x_3(t)) - b \cdot x_4(t) + d_0 \cdot d(t) \cdot \frac{l^2}{2} \cdot \cos(x_3(t)) \right) + \frac{1}{2} \cdot m \cdot l \cdot \cos(x_3(t)) \cdot \left(-\frac{1}{2} \cdot m \cdot l \cdot x_4(t)^2 \cdot \sin(x_3(t)) - c \cdot x_2(t) + \alpha_m \cdot i(t) - d_0 \cdot l \cdot d(t) \right)}{\frac{1}{3} \cdot m \cdot (M+m) \cdot l^2 - \frac{1}{4} \cdot m^2 \cdot l^2 \cdot \cos(x_3(t))^2} \end{bmatrix}
\end{aligned}$$

2.1. Response to initial conditions

The two graphs in Fig. 1 illustrate the response of the non-linear system to the following initial conditions: $x = 0$, $\dot{x} = 0$, $\theta = 1^\circ$ and $\dot{\theta} = 0$, obtained using the numerical solver `ode89`. As expected, the pendulum oscillates around the stable equilibrium position $\theta = 180^\circ$ under the effect of gravity acting on the rod. Regarding the cart dynamics, since no external horizontal forces act on the system, the total linear momentum is conserved throughout the motion.

Given that the system is initially at rest, the total momentum is zero, which implies that the horizontal position of the center of mass of the system remains constant at all times. As a consequence, when the pendulum swings in one direction, the cart must displace in the opposite direction to preserve the position of the center of mass, this results in the cart oscillating around its initial position $x = 0$, which is consistent with the simulation results. It should be noted that the center of mass is not exactly at $x = 0$ due to the initial angular displacement of the pendulum; however, since the horizontal projection of the pendulum displacement is proportional to $\sin(1^\circ) \approx 0.017$, this offset is considered negligible and it can be assumed that the center of mass remains at $x_{cm} \approx 0$ to a very good approximation. This is further confirmed by long-duration simulations, which reveal that the cart does not settle exactly at $x = 0$ but rather at an extremely small non-zero offset, consistent with the theoretical prediction that the true center of mass of the system is slightly displaced from the origin due to the initial angular condition. A small non-predicted behavior observed at the reversal points of the cart motion is discussed in Fig. 6.

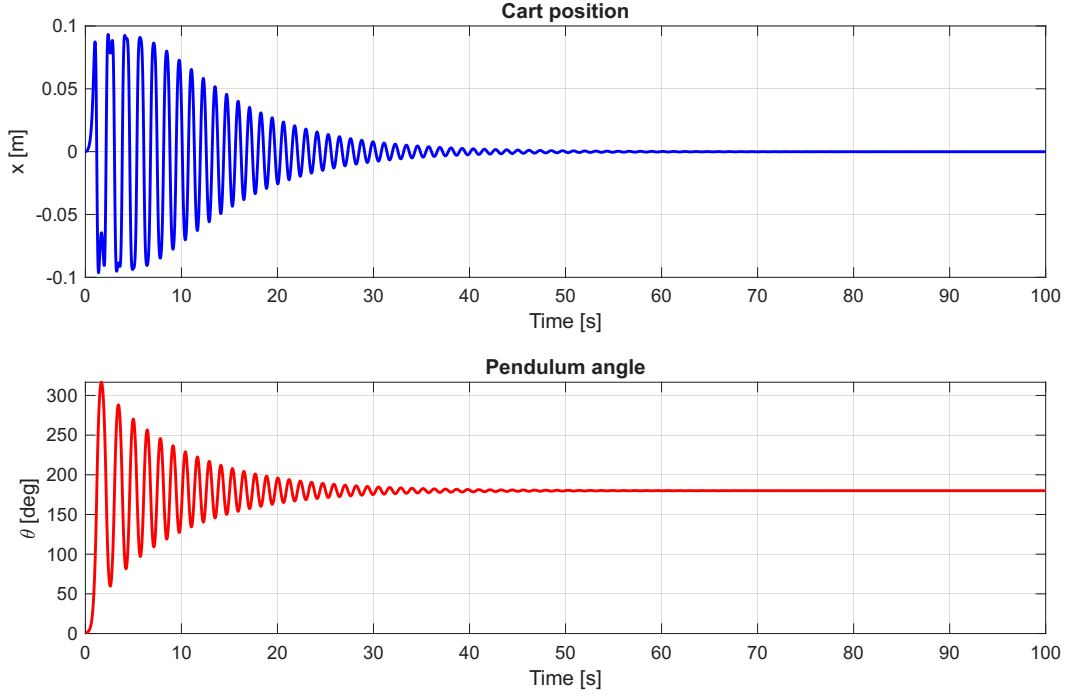


Figure 1: Response to initial conditions

The simulations were carried out using five different numerical integrators: the Forward and Backward Euler methods using a fixed step size of 0.01 s, and the built-in MATLAB adaptive step-size solvers `ode23`, `ode45`, and `ode89`, which employ embedded Runge-Kutta pairs of different orders to estimate the local error at each step and adjust the step size accordingly. When simulating over short time horizons, the solutions produced by the numerical solvers `ode23` and `ode89` are nearly identical, as illustrated in Fig. 2, where the differences remain negligible. However, the non-adaptive methods exhibit significant deviations from the expected behavior. The Forward Euler method tends to diverge for step sizes greater than 0.01s due to the low physical damping of the system, as its stability region does not encompass the imaginary axis, causing artificial energy accumulation. Conversely, the Backward Euler method introduces excessive numerical dissipation, resulting in a virtual damping that distorts the response in the final stages of the oscillation.

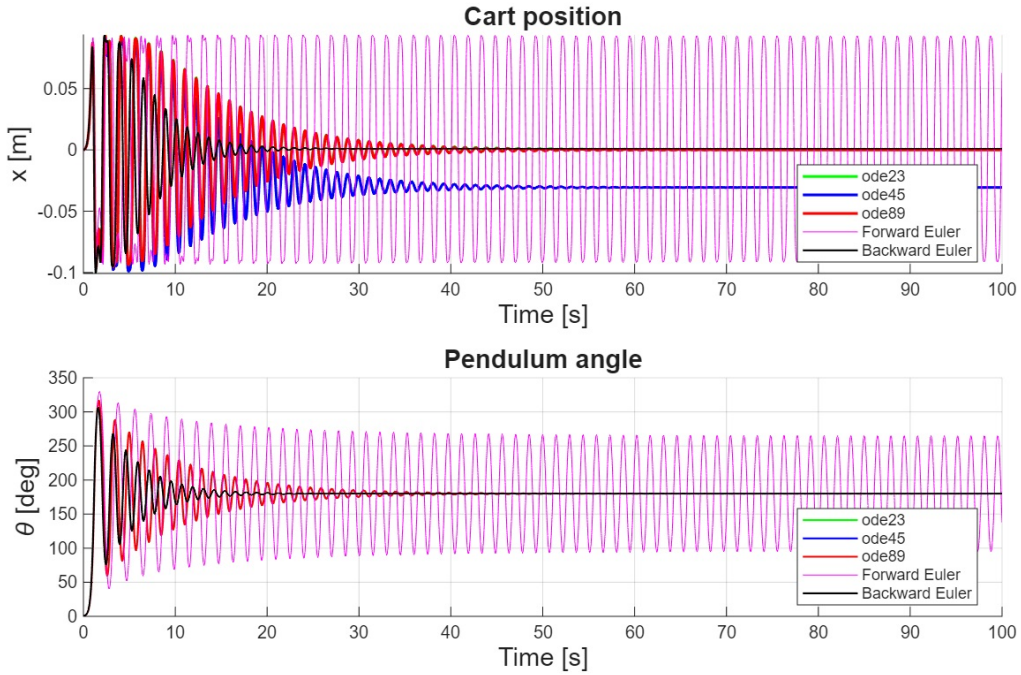


Figure 2: Comparison between the numerical solvers

Among the adaptive step-size methods, a notable difference emerges in the behavior of `ode45`, which fails to yield a cart position that converges to zero, as physically expected. This behavior can be attributed to the large step sizes selected by `ode45` throughout the simulation, as shown in Fig. 4, which leads to an accumulation of local errors sufficient to affect the long-term accuracy of the solution. When strict tolerances are imposed, such as an absolute tolerance of 10^{-12} and a relative tolerance of 10^{-6} , the step size is reduced accordingly and the method converges to zero, consistent with the results obtained by the other solvers.

A further relevant feature of adaptive solvers is observed in the step size evolution during the final stages of the simulation, shown in Figs. 3, 4, and 5. As the system approaches steady state, `ode89` maintains an approximately constant step size, which is physically consistent with the fact that the solution is no longer evolving significantly. In contrast, `ode45` and `ode23` exhibit a more irregular step size evolution, characterized by larger oscillations, which can be attributed to the lower order of the method and, consequently, less smooth operation of the adaptive step size controller.

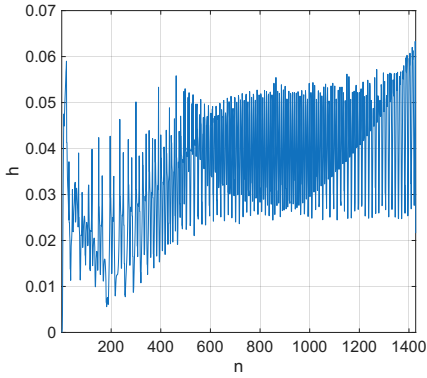


Figure 3: Step Size Ode23

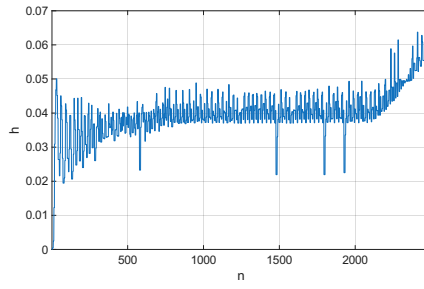


Figure 4: Step Size Ode45

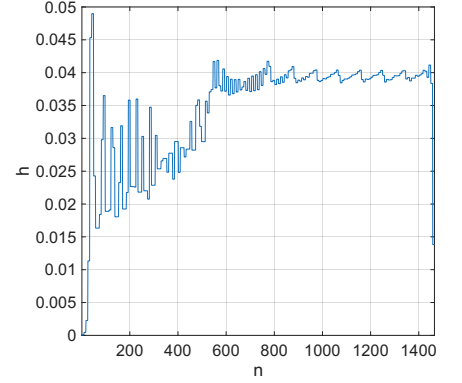


Figure 5: Step Size Ode89

As a final feature to evaluate, the computational time required to perform the simulations was analyzed. The execution times were recorded exclusively for the solvers `ode23`, `ode45`, and `ode89`, and the average times computed over multiple runs are $t_{\text{ode45}} = 0.208549$ s, $t_{\text{ode23}} = 0.102478$ s and $t_{\text{ode89}} = 0.106432$ s, including the initial simulations, which typically exhibit longer execution times. It is evident that `ode45` requires approximately twice the computational time compared to the other solvers, a consequence of the stricter tolerances necessary to achieve convergence, whereas `ode23` and `ode89` exhibit comparable execution times. Overall, for the simulation of the free response to initial conditions, `ode89` represents the most suitable choice, offering reduced computational cost alongside superior accuracy. This is particularly noteworthy given that `ode89` requires a higher number of function evaluations per step, a median of approximately 7000 evaluations compared to 4000 for `ode45`, yet compensates for this overhead by achieving convergence without the need for the tight tolerance constraints imposed on `ode45`.

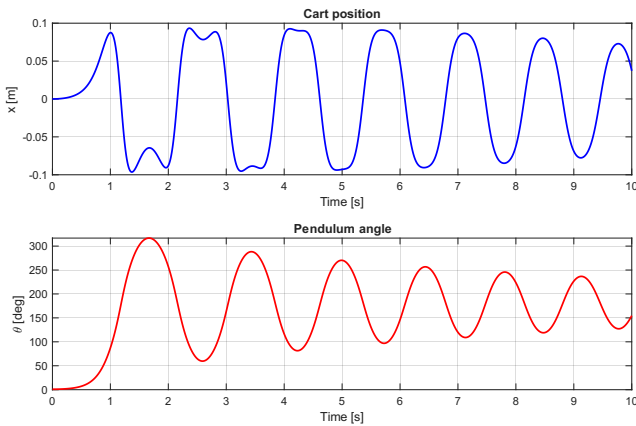


Figure 6: Non-intuitive feature

In general, the system does not exhibit any non-intuitive behavior; however, one phenomenon that seems unexpected is the brief oscillation of the cart near its reversal points. This effect, of negligibly small magnitude, can be attributed to the inertia of the pendulum as it swings in the vicinity of the unstable equilibrium point. Taking the first oscillation as a representative example, as the pendulum sweeps from $\theta = 0^\circ$ to $\theta = 270^\circ$, it drives the cart in the negative x -direction. Subsequently, as the pendulum continues from $\theta = 270^\circ$ toward $\theta = 360^\circ$, it imparts a force in the opposite direction; however, due to the limited kinetic energy available at this stage, the resulting cart displacement is minimal. The pendulum then falls back toward the equilibrium point, driving the cart in the positive x -direction. This phenomenon attenuates over time and becomes negligible after approximately 5 seconds, as the pendulum no longer retains sufficient energy to reach the $\theta = 270^\circ$ threshold.

2.2. Response to disturbance

The graphs below show the nonlinear open-loop response of the system to the disturbance obtained using the numerical solver `ode45` with default tolerances, over $t \in [0, 50]$ s, assuming zero initial conditions and a disturbance with rectangular time profile:

$$d(t) = \begin{cases} 0, & t < 1 \\ 1, & 1 \leq t \leq 3 \\ 0, & t > 3 \end{cases}$$

Regarding the relationship between the system response and the time profile of the disturbance, it is initially observed that the system's inertia delays the onset of the disturbance's effects. Furthermore, it is important to note that the impact of the disturbance increases proportionally with the exposed surface area of the rod. For instance, at $\theta = 90^\circ$ the disturbance has no effect; indeed, the system tends to drift towards positive x -direction despite the disturbance acting in the negative x -direction.

Generally, the system is observed to move more rapidly towards negative x values under the influence of the disturbance. Finally, no reduction in the amplitude of the θ oscillations is observed compared to the initial conditions. This occurs because the disturbance acts continuously: it decelerates the pendulum during its downward trajectory and accelerates it during its upward phase.

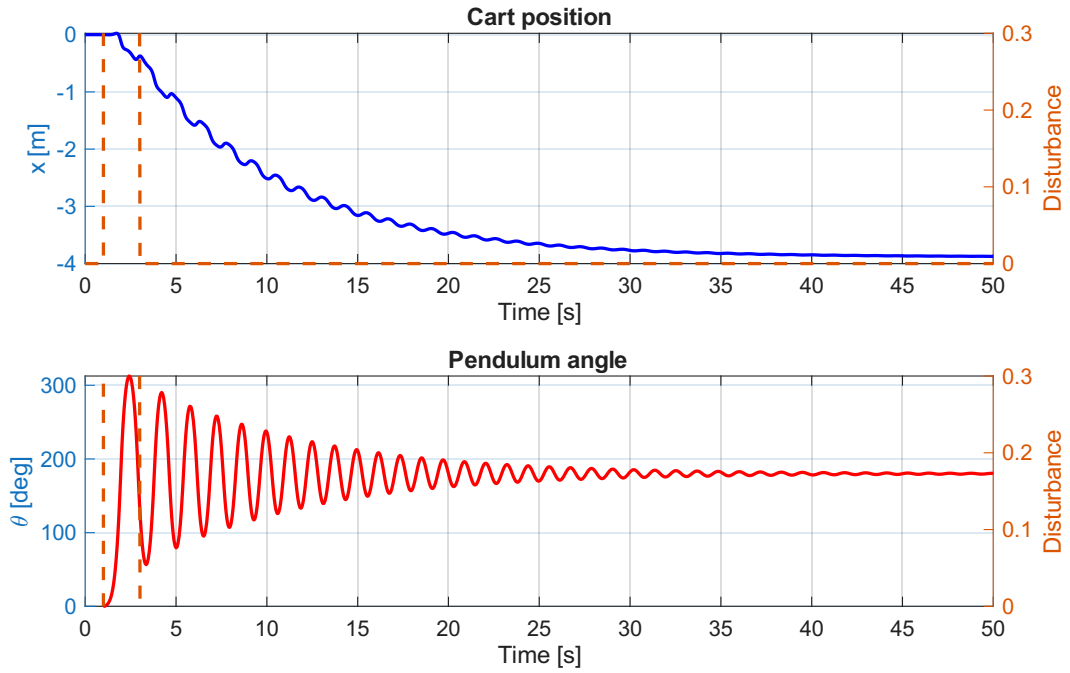


Figure 7: Time histories of $d(t)$, $x(t)$ and $\theta(t)$

Unlike the scenario described in Task 2.1, for this specific operating condition, `ode45` does not require strictly imposed tolerances to achieve convergence. Consequently, the choice of the optimal numerical solver differs. For all adaptive step size methods evaluated with default parameters (absolute tolerance of 10^{-6} and relative tolerance of 10^{-3}), the integration step size h initially assumes large values before stabilizing into smaller oscillatory patterns. Specifically, `ode23` exhibits the widest oscillations, ranging between 0.01 s and 0.09 s. In contrast, `ode45` oscillates within a narrower band (between 0.035 s and 0.046 s), while `ode89` maintains a relatively stable step size of approximately 0.036 s, as illustrated in Figs. 8, 9, and 10.

To evaluate the computational efficiency under these default conditions, the execution times of the three adaptive solvers were analyzed. The average computational times were $t_{\text{ode45}} = 0.024666$ s, $t_{\text{ode23}} = 0.033333$ s, and $t_{\text{ode89}} = 0.040666$ s.

These results highlight a reversal in performance compared to the previous analysis, with `ode45` emerging as the most efficient choice when tight constraints are not required.

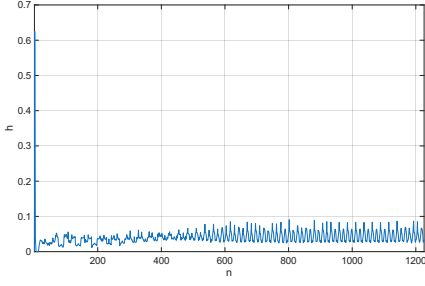


Figure 8: Step Size Ode23

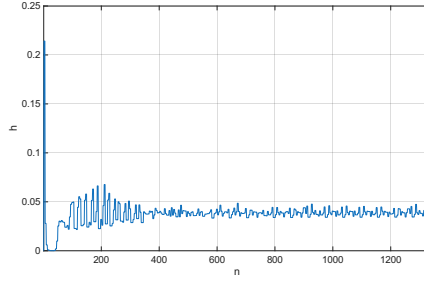


Figure 9: Step Size Ode45

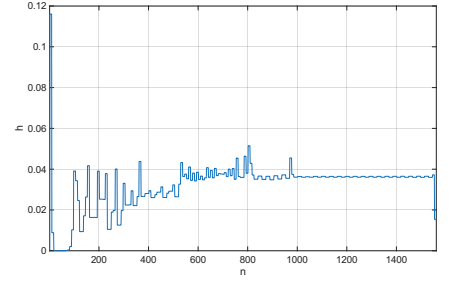


Figure 10: Step Size Ode89

Regarding the fixed-step solvers, the implementation of the Backward Euler method confirms its unconditional stability, consistently avoiding numerical divergence. However, this stability comes at a severe computational cost. As an implicit method applied to a nonlinear system, it necessitates the solution of a system of nonlinear algebraic equations at each time step. This overhead is clearly reflected in the recorded execution times: with an integration step of $h = 0.01$ s, the average simulation time was 0.7656 s. Reducing the step size by a factor of ten ($h = 0.001$ s) caused the execution time to surge to 7.6074 s. This significant increase stems directly from the iterative root-finding process required at every single time increment.

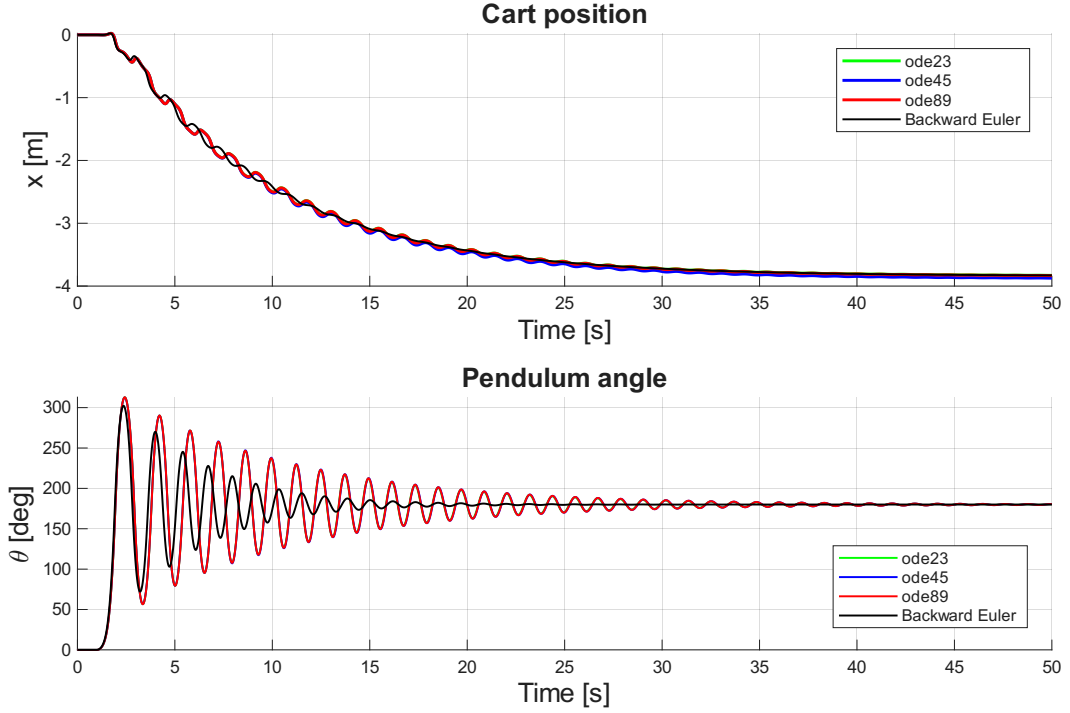


Figure 11: Comparison between the numerical solvers

Overall, according to the previous discussion and Fig. 11 for this specific system configuration and without the imposition of strict tolerances, **ode45** represents the most computationally efficient solver, offering an optimal balance between execution speed and integration accuracy.

2.3. Discussion

The first scenario (Task 2.1) evaluates the open-loop system's natural energy dissipation and equilibrium-seeking behavior. Initiated with a 1° pendulum angle θ , the system exhibits damped oscillations due to friction, preventing a full rotation and causing the pendulum to settle at its stable, downward equilibrium of 180° . Concurrently, the cart position x oscillates around its initial zero position before coming to rest. This observed response aligns perfectly with the physical expectations of the system. The cart's return to the origin demonstrates that the internal reaction forces balance out symmetrically during the damped oscillations, with no net external horizontal translation. Additionally, the system clearly shows dynamic coupling: the rightward swing of the pendulum induces a leftward reaction force on the cart, and vice versa, keeping the overall horizontal displacement centered.

In contrast, Task 2.2 shifts the focus from an initial condition to an external perturbation. In this scenario, the disturbance is introduced between $t = 1$ s and $t = 3$ s. Looking at the time histories, the cart position x is significantly affected by the horizontal force acting towards the left. During this window, the natural oscillatory motion of the cart is largely overridden by the external forcing. On the other hand, the angle θ is affected by the disturbance, but its impact remains relatively weak compared to the massive displacement of the cart. This behavior aligns completely with the expected dynamics of the system. Rather than smoothly gliding, the cart experiences a macroscopic translation to the left while continuing to oscillate, as the forced leftward movement is superimposed with the reaction forces of the swinging pendulum. After the disturbance is removed at $t = 3$ s, the cart continues to move left while oscillating. Friction progressively dissipates the remaining mechanical energy, causing the cart to ultimately settle at a steady-state position of $x = -4$ m.

In conclusion, the time histories of both scenarios exhibit physically consistent behavior. The non-linear simulation accurately captures the system's dynamic mass-inertia coupling and energy dissipation mechanisms, validating its representation of the real-world system.

3. Nonlinear Analysis and Simulation (Simulink model)

A Simulink model was developed to simulate the nonlinear response of the system to the disturbance defined in Task 2.2, assuming zero initial condition.

3.1. Response to disturbance

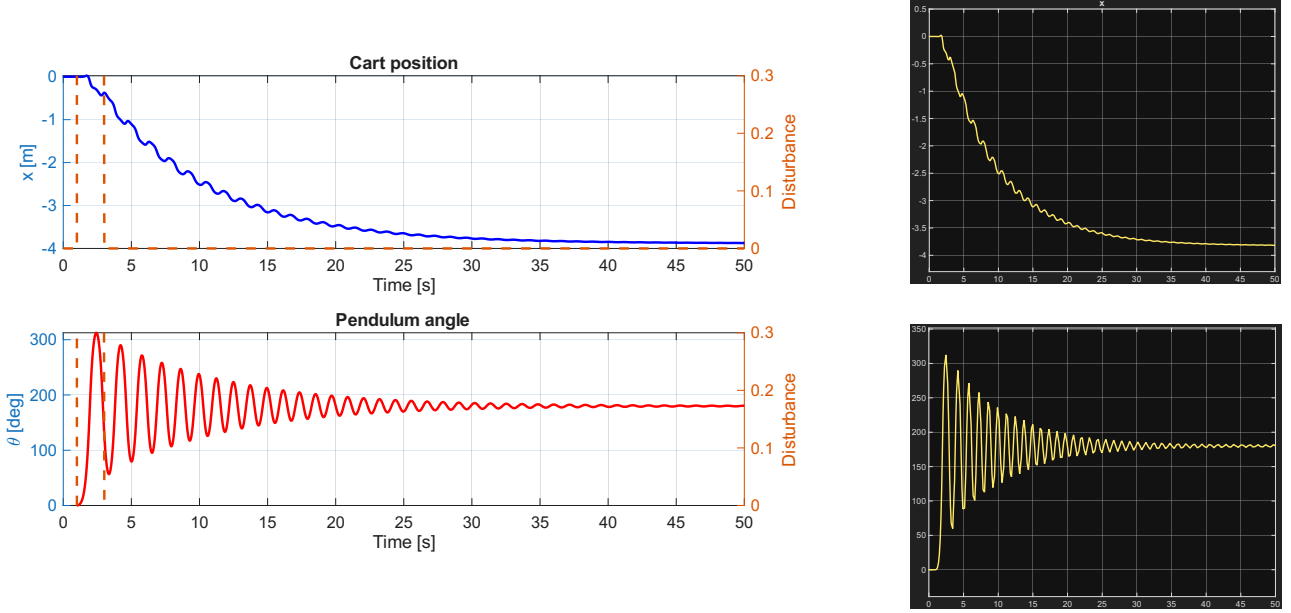


Figure 12: Response to the disturbance (Ode45), Matlab on the left, Simulink on the right

The time histories for the external disturbance $d(t)$, cart position $x(t)$, and pendulum angle $\theta(t)$ derived from the Simulink model were directly compared against the MATLAB script output. The comparison demonstrates a perfect overlap between the two responses, with identical dynamic behavior throughout both the transient disturbance phase and the subsequent stabilization. This confirms the fundamental correctness of the non-linear differential equation implementations across both environments. Any microscopic discrepancies are negligible and inherently attributable to solver-specific numerical integration tolerances and dynamic time-stepping differences between the Simulink engine and the MATLAB solver.

The most significant difference observed is the computational time. In the examined case (Fig. 12), using the `ode45` solver (with default tolerances), the computational time of the MATLAB script is 0.025 s, whereas the Simulink model takes 0.3284 s. This highlights that running the simulation through the Simulink engine, especially when utilizing a MATLAB Function block inside the model, which introduces additional execution overhead, is significantly slower than executing a pure MATLAB script.

3.2. Discussion

The selection of the numerical solver for the Simulink model was justified through an empirical comparison evaluating both numerical stability and computational efficiency. As illustrated in Fig. 13, applying a fixed-step Forward Euler solver (`ode1`) with a step size of $h = 0.01$ s causes system divergence, as this step size exceeds the stability boundaries of the first-order explicit method for this specific dynamic system. Achieving stability required reducing the step size to $h = 0.001$ s, which prevented divergence but imposed a severe computational penalty, yielding the highest execution time (0.4228 s).

In contrast, variable-step solvers (`ode23` and `ode45`) optimize computational effort by utilizing small time-steps during rapid disturbance transients and larger steps near equilibrium. Notably, the time histories reveal that the pendulum angle simulated via `ode23` appears to settle at the $\theta = 180^\circ$ equilibrium slightly faster than with `ode45` (as shown in Figs. 14 and 15).

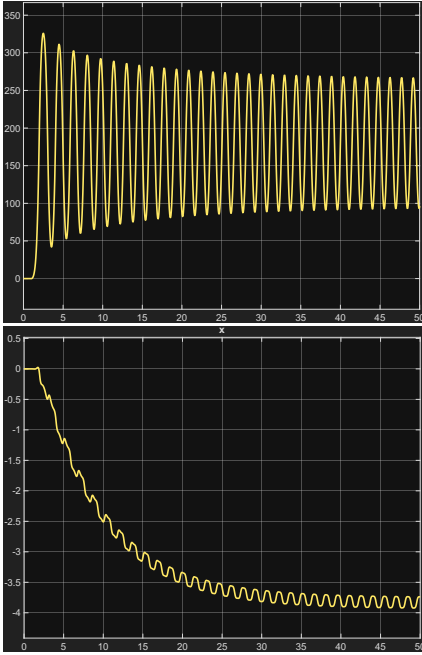


Figure 13: Euler $h = 0.01$

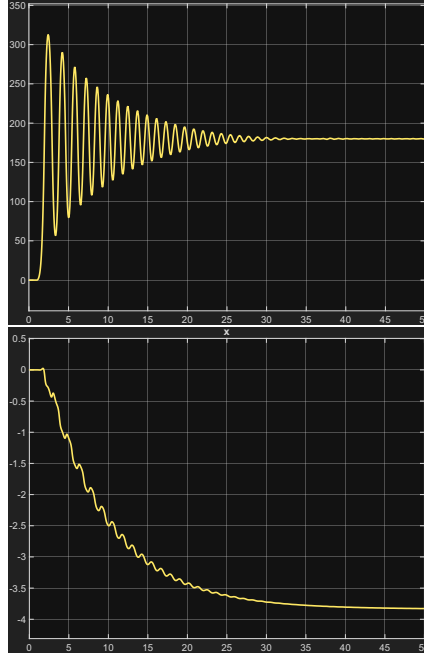


Figure 14: Ode 23 with disturbance

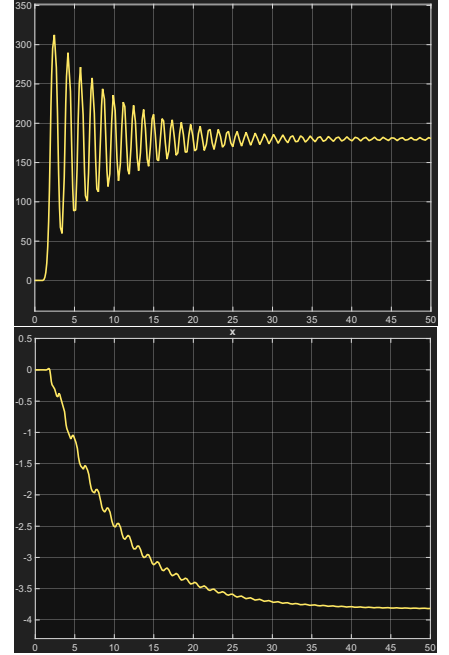


Figure 15: Ode 45 with disturbance

This discrepancy is not an indicator of superior solver performance; rather, it highlights the numerical damping inherent in lower-order integration schemes. As the system nears equilibrium and oscillation amplitudes become microscopic, the fourth/fifth-order `ode45` accurately resolves these minute residual dynamics, inherently taking longer to settle strictly within tolerance limits. The second/third-order `ode23`, however, incurs a higher truncation error per step that artificially dampens the system. This prematurely eliminates high-frequency micro-oscillations, falsely presenting a faster settling time. This phenomenon explicitly validates the selection of `ode45` as the optimal solver, as it maintains high dynamic fidelity without introducing artificial, numerical dissipation to the physical model.

4. Linear Analysis and Simulation

Based on the nonlinear equations derived in (3), the system is linearized around the unstable upright equilibrium configuration $\theta_{eq} = 0$, $\dot{\theta}_{eq} = 0$, $\ddot{\theta}_{eq} = 0$. Note that the independent coordinate x and its derivatives are not constrained to zero, allowing for a generic equilibrium configuration.

4.1. Linear model

Linearization is performed using a first-order Taylor series expansion with respect to all variables. The independent coordinate x and its time derivatives are denoted as x_{eq} , $\dot{x}_{eq}$, $\ddot{x}_{eq}$.

$$\tilde{f} = f(\theta_{eq}, 0, 0, x_{eq}, \dot{x}_{eq}, \ddot{x}_{eq}) + \frac{\partial f}{\partial \theta} \cdot \Delta\theta + \frac{\partial f}{\partial \dot{\theta}} \cdot \Delta\dot{\theta} + \frac{\partial f}{\partial \ddot{\theta}} \cdot \Delta\ddot{\theta} + \frac{\partial f}{\partial x} \cdot \Delta x + \frac{\partial f}{\partial \dot{x}} \cdot \Delta\dot{x} + \frac{\partial f}{\partial \ddot{x}} \cdot \Delta\ddot{x} \quad (19)$$

where $\Delta x = x - x_{eq}$, $\Delta\theta = \theta - \theta_{eq}$ and their derivatives are defined analogously.

Referring to the first equation of (3) as f_1

$$\begin{aligned} \tilde{f}_1 &= -I_1 \cdot \ddot{x}_{eq} - \alpha_1 \cdot d(t) - g \cdot I_1 \cdot \theta + b \cdot \dot{\theta} + I_2 \cdot \ddot{\theta} - I_1 \cdot \Delta\ddot{x} = 0 \\ &= I_2 \cdot \ddot{\theta} + b \cdot \dot{\theta} - I_1 \cdot g \cdot \theta - I_1 \cdot \ddot{x} - \alpha_1 \cdot d(t) = 0 \\ &\Rightarrow I_2 \cdot \ddot{\theta} + b \cdot \dot{\theta} - I_1 \cdot g \cdot \theta - I_1 \cdot \ddot{x} = \alpha_1 \cdot d(t) \end{aligned} \quad (20)$$

Referring to the second equation of (3) as f_2

$$\begin{aligned} \tilde{f}_2 &= (M + I_0) \cdot \ddot{x}_{eq} + c \cdot \dot{x}_{eq} - \alpha_m \cdot i(t) + \alpha_0 \cdot d(t) - I_1 \cdot \ddot{\theta} + c \cdot \Delta\dot{x} + (M + I_0) \cdot \Delta\ddot{x} = 0 \\ &= (M + I_0) \cdot \ddot{x} + c \cdot \dot{x} - I_1 \cdot \ddot{\theta} - \alpha_m \cdot i(t) + \alpha_0 \cdot d(t) = 0 \\ &\Rightarrow (M + I_0) \cdot \ddot{x} + c \cdot \dot{x} - I_1 \cdot \ddot{\theta} = \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) \end{aligned} \quad (21)$$

Performing the linearization at the origin ($x_{eq} = 0$, $\dot{x}_{eq} = 0$, $\ddot{x}_{eq} = 0$) yields the exact same results. Physically, this is because the forces acting on the system do not depend on the absolute position of the cart (non-positional forces). Consequently, the resulting linearized equations are independent of the specific reference position x_{eq} .

In order to obtain the explicit form of the linear equation it is useful to use the matrix form:

$$\begin{bmatrix} I_2 & -I_1 \\ -I_1 & (M + I_0) \end{bmatrix} \cdot \begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} \alpha_1 \cdot d(t) + I_1 \cdot g \cdot \theta - b \cdot \dot{\theta} \\ \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) - c \cdot \dot{x} \end{bmatrix} \quad (22)$$

For simplicity, the right-hand side term of (22) is denoted by the vector $\begin{bmatrix} D_1 \\ D_2 \end{bmatrix}$

$$\begin{bmatrix} \ddot{\theta} \\ \ddot{x} \end{bmatrix} = \frac{1}{I_2 \cdot (M + I_0) - I_1^2} \cdot \begin{bmatrix} (M + I_0) & I_1 \\ I_1 & I_2 \end{bmatrix} \cdot \begin{bmatrix} D_1 \\ D_2 \end{bmatrix} \quad (23)$$

$$\begin{cases} \ddot{\theta} = \frac{(M + I_0) \cdot D_1 + I_1 \cdot D_2}{I_2 \cdot (M + I_0) - I_1^2} \\ \ddot{x} = \frac{I_1 \cdot D_1 + I_2 \cdot D_2}{I_2 \cdot (M + I_0) - I_1^2} \end{cases}$$

Finally, the state-space representation of the system is derived, yielding the matrices A, B, C, and D

$$\vec{z} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \rightarrow \vec{\dot{z}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} x_2 \\ \frac{I_1 \cdot D_1 + I_2 \cdot D_2}{I_2 \cdot (M + I_0) - I_1^2} \\ x_4 \\ \frac{(M + I_0) \cdot D_1 + I_1 \cdot D_2}{I_2 \cdot (M + I_0) - I_1^2} \end{bmatrix}$$

The determinant of the inertia matrix of (22) is denoted by Δ

$$\ddot{\vec{z}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-c \cdot I_2}{\Delta} & \frac{I_1^2 \cdot g}{\Delta} & \frac{-b \cdot I_1}{\Delta} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{-c \cdot I_1}{\Delta} & \frac{(M + I_0) \cdot I_1 \cdot g}{\Delta} & \frac{-(M + I_0) \cdot b}{\Delta} \end{bmatrix}}_A \cdot \vec{z} + \underbrace{\begin{bmatrix} 0 \\ \frac{I_2 \cdot \alpha_m}{\Delta} \\ 0 \\ \frac{I_1 \cdot \alpha_m}{\Delta} \end{bmatrix}}_{B_i} \cdot i(t) + \underbrace{\begin{bmatrix} 0 \\ \frac{I_1 \cdot \alpha_1 - I_2 \cdot \alpha_0}{\Delta} \\ 0 \\ \frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{\Delta} \end{bmatrix}}_{B_d} \cdot d(t) \quad (24)$$

$$\ddot{\vec{y}} = \begin{bmatrix} x(t) \\ \theta(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_C \cdot \vec{z} + \underbrace{0}_D \cdot \begin{bmatrix} i(t) \\ d(t) \end{bmatrix} \quad (25)$$

4.2. Response to initial conditions and comparison with the nonlinear model

Fig. 16 shows the response of the linearized system, obtained by linearizing around the unstable equilibrium $\theta = 0$, to the initial conditions $x = 0$, $\dot{x} = 0$, $\theta = 1^\circ$ and $\dot{\theta} = 0$. The response is reported only up to $t = 1.8$ s, as the system departs rapidly from the linearization point and, beyond this interval, the linearized model loses its physical validity: the small-angle assumptions upon which it is based are no longer satisfied, and the solution ceases to be representative of the true non-linear behavior (as illustrated in Figs. 17 and 18).

The discrepancy between the linearized and non-linear responses can be attributed to several factors. First, the non-linear equations of motion contain the term $\dot{\theta}^2$, which vanishes upon linearization. For sufficiently large angular displacements, the rotational velocity becomes non-negligible and the centrifugal force of the pendulum exerts a significant influence on the dynamics, an effect that is entirely absent in the linear model. Second, the linearization replaces $\sin(\theta)$ with θ , introducing an error in the gravitational force. Since $\sin(\theta) < \theta$ for all $\theta > 0$, the linearized model overestimates the gravitational force acting on the pendulum, causing the linear solution to diverge more rapidly than the true non-linear response. Taken together, these approximations define a finite time interval within which the linearized model provides an accurate description of the system dynamics, beyond which the non-linear terms become dominant and the linear approximation is no longer valid.

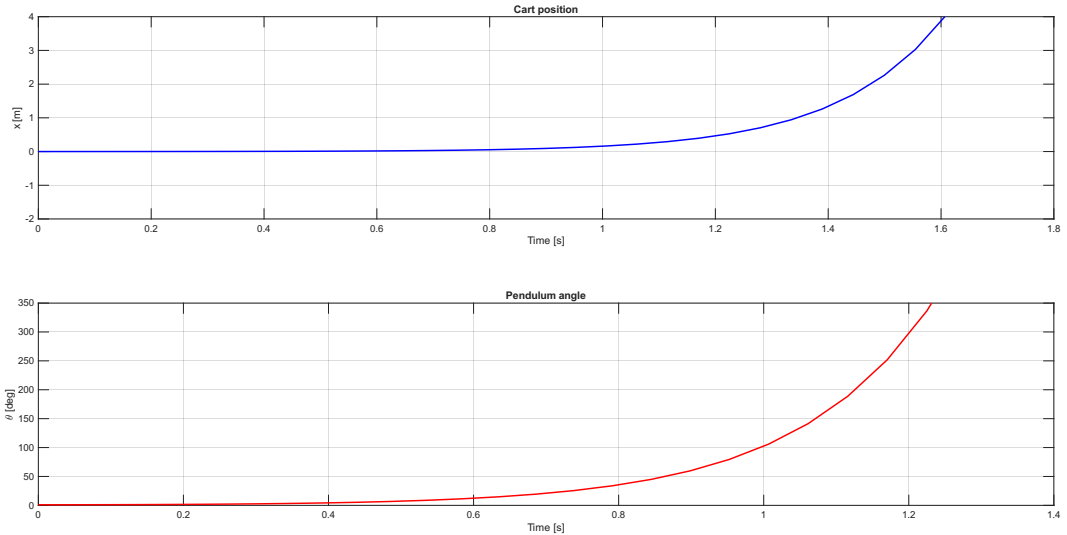


Figure 16: Linear model response to initial condition

In order to evaluate the valid time interval for the linear model, several methods exist. The standard approach is based on the small-angle approximation criterion: the linearization assumes $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$, which introduces a relative error of approximately 5% at $\theta \approx 17^\circ$, representing the conventional threshold beyond which the linear model can no longer be considered accurate. From the simulation results, this angle is reached at approximately $t \approx 0.7$ s, which is therefore taken as the upper bound of the validity interval of the linearized model (also confirmed by the comparisons in Figs. 17 and 18).

An alternative approach involves directly comparing the linear and nonlinear solutions and evaluating the time at which their discrepancy exceeds a tolerance. This method is much more stringent due to the accumulation of errors from approximations beyond just the small-angle assumption.

A further criterion may be formulated in terms of the control effort required to stabilize the pendulum. Since the controller is designed on the basis of the linearized model, its performance is inherently tied to the validity of the linear approximation. When the discrepancy between the control effort predicted by the linear model and that required by the true non-linear system exceeds a prescribed threshold, the linear model can no longer be considered a reliable basis for control design.

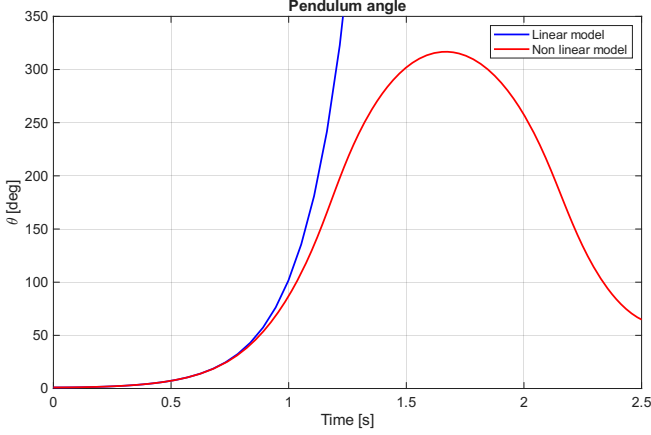


Figure 17: θ comparison between linear and non-linear

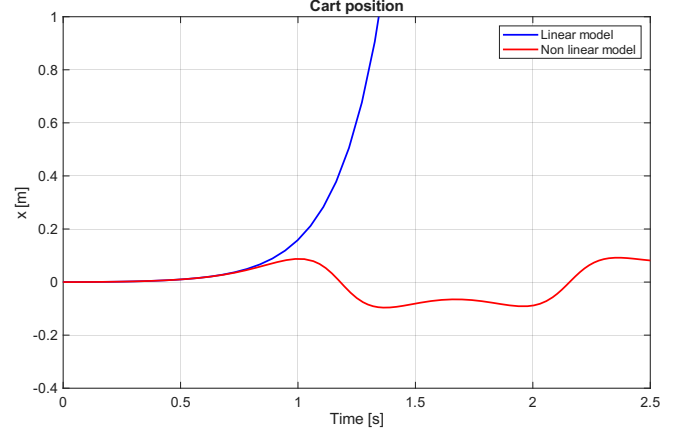


Figure 18: x comparison between linear and non-linear

A notable feature observed when studying the linearized system is the discrepancy that arises between two simulation approaches: numerical integration via `ode45` applied to the linearized equations, and the built-in MATLAB command `initial` applied to the state-space representation. The differences are not substantial in terms of physical accuracy (both methods converge to the same solution) but rather manifest in how each solver determines the points at which the solution is evaluated.

Because `ode45` employs a variable-step Runge-Kutta scheme, the solver tends to take larger steps where the solution varies slowly, producing a non-uniform and relatively sparse set of evaluation points. This leads to a solution that, when plotted, appears less smooth due to the coarser sampling in regions of slow dynamics.

In contrast, the `initial` command computes the free response of a state-space system analytically, exploiting the closed-form solution of the linear time-invariant system. This is based on the matrix exponential solution

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}_0 \quad (26)$$

where the temporal evolution is entirely governed by the eigenstructure of the system matrix $\mathbf{A}$. The command internally analyses the eigenvalues of $\mathbf{A}$ to characterize the dominant timescales of the system, such as natural frequencies and decay rates, and selects a uniform time grid whose density is sufficient to resolve the fastest dynamic present. The result is a denser, regularly spaced set of evaluation points, yielding a visually smoother output.

The central role of the eigenvalues can be understood by recalling that the general solution of the linear system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is constructed from the eigenvalues λ_i and eigenvectors $\mathbf{v}_i$ of $\mathbf{A}$, with each eigenvalue contributing a natural mode of the form:

$$\mathbf{x}_i(t) = c_i \mathbf{v}_i e^{\lambda_i t} \quad (27)$$

For complex eigenvalues $\lambda = \sigma \pm j\omega$, the exponential term expands as:

$$e^{\lambda t} = e^{\sigma t} (\cos(\omega t) + j \sin(\omega t)) \quad (28)$$

where the real part σ governs the rate of damping or growth, and the imaginary part ω determines the frequency of oscillation. The `initial` command exploits this structure to identify the fastest mode present in the system, that is, the eigenvalue with the largest magnitude $|\lambda_i|$, which in turn imposes the minimum temporal resolution required to accurately represent

the solution. Sampling at a rate insufficient to resolve $\omega_{\max}$ would result in aliasing or loss of the high-frequency oscillatory content, analogously to the Nyquist criterion in signal theory. The time grid is therefore chosen such that Δt is small enough to capture all modes present. By contrast, `ode45` has no a priori knowledge of the system's eigenstructure and discovers the necessary resolution adaptively through local error control during integration, which explains why it may use considerably fewer evaluation points in regions where the solution evolves slowly.

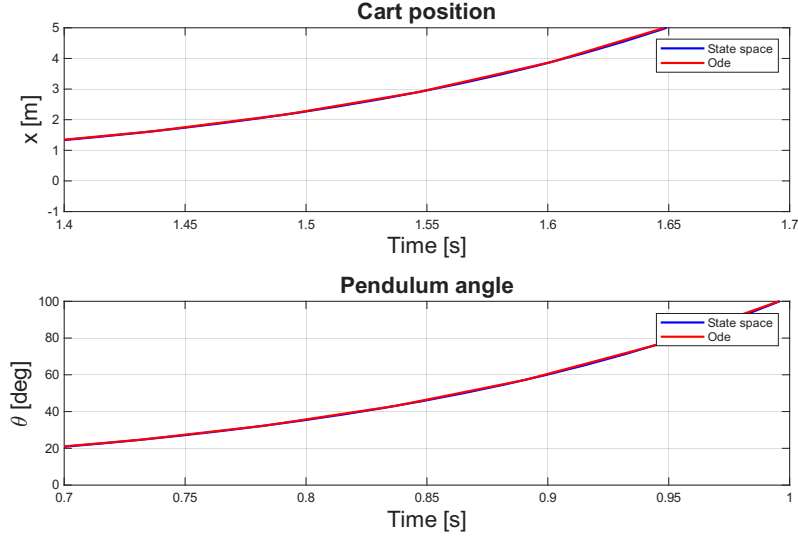


Figure 19: Comparison between ode and state-space model

4.3. Frictionless linear model

The frictionless linear model is derived from (20) and (21) by setting the friction coefficients c and b equal to zero. The resulting equations read:

$$\begin{cases} -I_1 \cdot \ddot{x} + I_2 \cdot \ddot{\theta} - I_1 \cdot g \cdot \theta = \alpha_1 \cdot d(t) \\ (M + I_0) \cdot \ddot{x} - I_1 \cdot \ddot{\theta} = \alpha_m \cdot i(t) - \alpha_0 \cdot d(t) \end{cases} \quad (29)$$

And the state-space model is:

$$\vec{z} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \rightarrow \vec{\dot{z}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$$

The determinant of the inertia matrix is the same of (22) and again it is denoted by Δ

$$\vec{\dot{z}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{I_1^2 \cdot g}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M + I_0) \cdot I_1 \cdot g}{\Delta} & 0 \end{bmatrix}}_A \cdot \vec{z} + \underbrace{\begin{bmatrix} 0 \\ \frac{I_2 \cdot \alpha_m}{\Delta} \\ 0 \\ \frac{I_1 \cdot \alpha_m}{\Delta} \end{bmatrix}}_{B_i} \cdot i(t) + \underbrace{\begin{bmatrix} 0 \\ \frac{I_1 \cdot \alpha_1 - I_2 \cdot \alpha_0}{\Delta} \\ 0 \\ \frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{\Delta} \end{bmatrix}}_{B_d} \cdot d(t) \quad (30)$$

$$\vec{y} = \begin{bmatrix} x(t) \\ \theta(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_C \cdot \vec{z} + \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_D \cdot \begin{bmatrix} i(t) \\ d(t) \end{bmatrix} \quad (31)$$

4.4. Transfer functions

By applying the *Laplace transform*, the governing equations are cast into the frequency domain, from which the transfer functions are subsequently obtained.

$$\begin{cases} -I_1 \cdot s^2 \cdot X(s) + (I_2 \cdot s^2 - I_1 \cdot g) \cdot \vartheta(s) = \alpha_1 \cdot D(s) \\ (M + I_0) \cdot s^2 \cdot X(s) - I_1 \cdot s^2 \cdot \vartheta(s) = \alpha_m \cdot I(s) - \alpha_0 \cdot D(s) \end{cases} \quad (32)$$

Using the the matrix form and deriving the explicit form

$$\begin{aligned} \begin{bmatrix} I_1 \cdot s^2 & (I_2 \cdot s^2 - I_1 \cdot g) \\ (M + I_0) \cdot s^2 & -I_1 \cdot s^2 \end{bmatrix} \cdot \begin{bmatrix} X(s) \\ \vartheta(s) \end{bmatrix} &= \begin{bmatrix} \alpha_1 \cdot D(s) \\ \alpha_m \cdot I(s) - \alpha_0 \cdot D(s) \end{bmatrix} \\ \begin{bmatrix} X(s) \\ \vartheta(s) \end{bmatrix} &= \frac{1}{s^2 \cdot [I_1^2 \cdot s^2 - (M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g)]} \cdot \begin{bmatrix} -I_1 \cdot s^2 & -(I_2 \cdot s^2 - I_1 \cdot g) \\ -(M + I_0) \cdot s^2 & -I_1 \cdot s^2 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \cdot D(s) \\ \alpha_m \cdot I(s) - \alpha_0 \cdot D(s) \end{bmatrix} \\ \Rightarrow X(s) &= \frac{-I_1 \cdot s^2 \cdot \alpha_1 \cdot D(s) - (I_2 \cdot s^2 - I_1 \cdot g) [\alpha_m \cdot I(s) - \alpha_0 \cdot D(s)]}{s^2 \cdot [I_1^2 \cdot s^2 - (M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g)]} \\ \Rightarrow \vartheta(s) &= \frac{-(M + I_0) \cdot s^2 \cdot \alpha_1 \cdot D(s) - I_1 \cdot s^2 \cdot [\alpha_m \cdot I(s) - \alpha_0 \cdot D(s)]}{s^2 \cdot [I_1^2 \cdot s^2 - (M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g)]} \end{aligned} \quad (33)$$

In order to obtain the transfer functions the non-used input is imposed as equal to zero

$$\begin{aligned} G_{\vartheta, I}(s) &= \frac{\vartheta(s)}{I(s)} = \frac{I_1 \cdot s^2 \cdot \alpha_m}{s^2 \cdot [(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} = \frac{I_1 \cdot \alpha_m}{[(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} = \frac{\frac{I_1 \cdot \alpha_m}{(M + I_0) \cdot I_2 - I_1^2}}{s^2 - \frac{I_1 \cdot g \cdot (M + I_0)}{(M + I_0) \cdot I_2 - I_1^2}} \\ G_{\vartheta, D}(s) &= \frac{\vartheta(s)}{D(s)} = \frac{(M + I_0) \cdot s^2 \cdot \alpha_1 - I_1 \cdot s^2 \cdot \alpha_0}{s^2 \cdot [(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} = \frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{[(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} = \frac{\frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{(M + I_0) \cdot I_2 - I_1^2}}{s^2 - \frac{I_1 \cdot g \cdot (M + I_0)}{(M + I_0) \cdot I_2 - I_1^2}} \\ G_{X, I}(s) &= \frac{X(s)}{I(s)} = \frac{(I_2 \cdot s^2 - I_1 \cdot g) \cdot \alpha_m}{s^2 \cdot [(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} \\ G_{X, D}(s) &= \frac{X(s)}{D(s)} = \frac{I_1 \cdot s^2 \cdot \alpha_1 - (I_2 \cdot s^2 - I_1 \cdot g) \cdot \alpha_0}{s^2 \cdot [(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} \end{aligned}$$

4.5. Pole-zero analysis

The graphs in Fig. 20 show the position of the poles and the zeros of the four transfer functions derived in subsection 4.4. As a preliminary observation, every function presents a real positive pole which leads to the instability of every free coordinate in respect to every input. Considering the transfer functions $G_{\vartheta, I}$ and $G_{\vartheta, D}$, both share the same pole configuration, with one pole located in the right-half plane and one in the left-half plane symmetrically placed with respect to the imaginary axis, as visible in Fig. 20. The poles of a transfer function correspond exactly to the eigenvalues of the system matrix $\mathbf{A}$, since the denominator of $G(s)$ is the characteristic polynomial $\det(s\mathbf{I} - \mathbf{A})$, whose roots are simultaneously the poles of $G(s)$ and the eigenvalues of $\mathbf{A}$. It is important to note that, in a coupled multi-degree-of-freedom system, the eigenvalues cannot be strictly partitioned into those governing the dynamics of θ and those governing the dynamics of x : each eigenvalue and its associated eigenvector characterizes a natural mode of the global system, in which all state variables participate simultaneously.

Nevertheless, this pole pair strongly associates with the pendulum's rotational dynamics, reflecting the nonlinear system's two equilibria: asymptotically stable ($\theta = 180^\circ$) and unstable ($\theta = 0^\circ$). The linearization about the unstable equilibrium preserves this duality through the symmetric pole configuration characteristic of a saddle point: one mode decays exponentially, reflecting the stable manifold (the ideal trajectory converging exactly to 0°), while the other grows unboundedly, reflecting the repulsive nature and instability of $\theta = 0^\circ$.

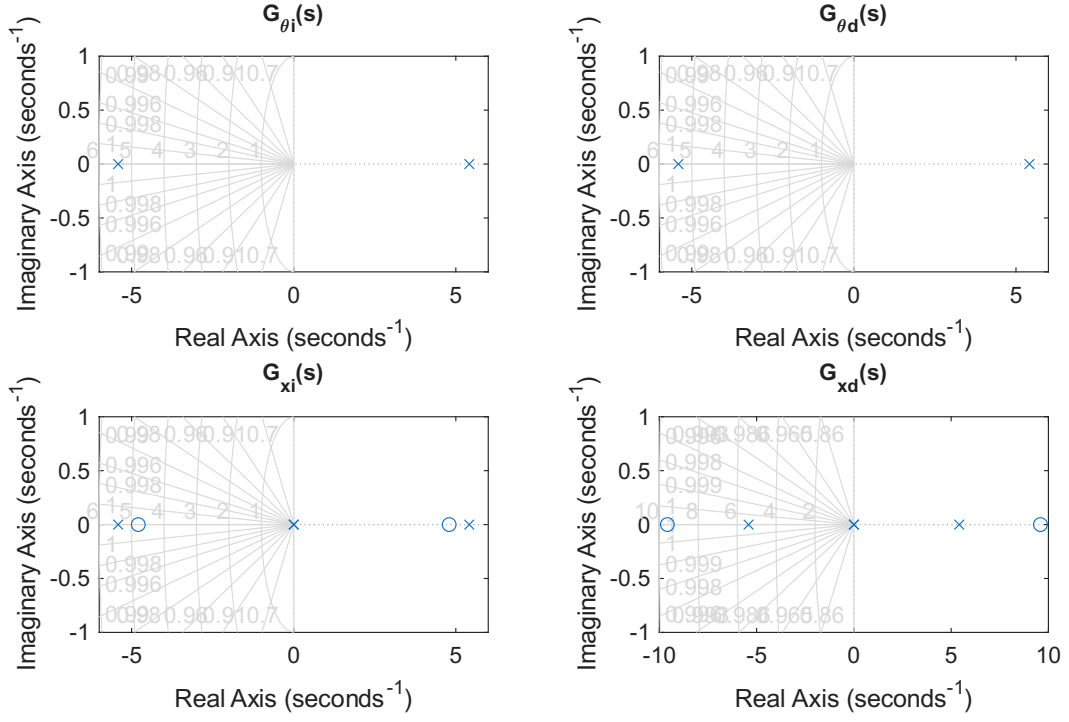


Figure 20: Pole-zero map of the four transfer functions

Considering the transfer functions $G_{x,I}(s)$ and $G_{x,D}(s)$, both share the same pole configuration as $G_{\vartheta,I}(s)$ and $G_{\vartheta,D}(s)$, since the poles are the eigenvalues of the system matrix $\mathbf{A}$. In particular, both present a positive real pole, driving the instability of the x coordinate with respect to both inputs. Additionally, a pole at the origin $s = 0$ is present: rather than decaying or growing exponentially, this mode causes the response to drift linearly in time.

Unlike $G_{\vartheta,I}(s)$ and $G_{\vartheta,D}(s)$, the two transfer functions $G_{x,I}(s)$ and $G_{x,D}(s)$ are not identical to each other, presenting different zero locations. This is expected: while the poles are determined solely by $\mathbf{A}$ and are therefore common to all transfer functions, the zeros depend on the specific input and output considered, being determined by the $\mathbf{B}$ and $\mathbf{C}$ matrices. Both transfer functions present zeros in both the left-half and the right-half of the complex plane. The presence of a right-half plane zero classifies both systems as *non-minimum phase*: for a step input, which is representative of both the current input and the disturbance considered in this study, the response of x will initially move in the direction opposite to the expected one before eventually diverging due to the unstable pole. The main difference between $G_{x,I}(s)$ and $G_{x,D}(s)$ lies in the location of the right-half-plane zero, which is positioned further from the origin in $G_{x,D}(s)$, implying that the non-minimum phase behavior manifests on a different timescale in the two cases.

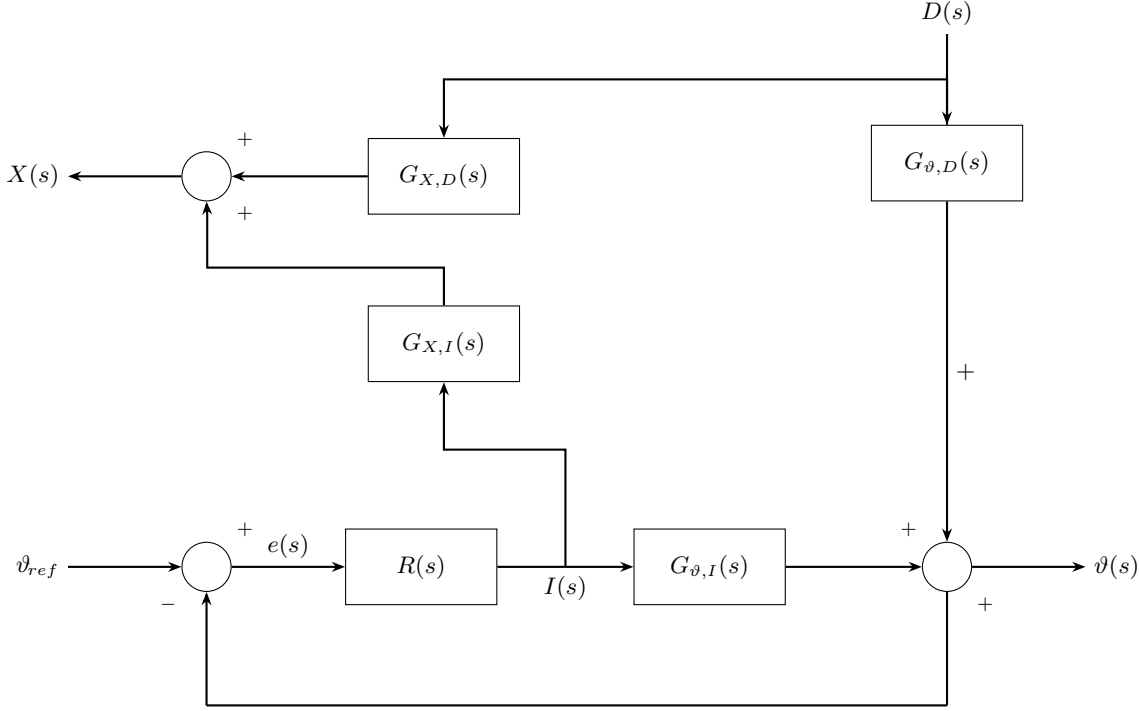
5. Classical Feedback Control

This task presents the design of two distinct regulators for the linearized frictionless system. Specifically, a PD and a PID controller are developed to regulate the pendulum angle.

The subsequent sections will highlight the limitations of both controllers, analyzing their performance in terms of rod inclination, cart position, and motor current.

5.1. Closed-loop block diagram

This section provides the block diagram used for the pendulum angle control. The Single-Input Single-Output (SISO) feedback loop focused solely on the angle θ is clearly visible. Furthermore, it is apparent that no control action is implemented for the cart position.



5.2. Closed-loop model (PD controller)

In general, a Proportional-Derivative (PD) controller is expressed in the form of: $R(s) = K_p + K_d \cdot s$. The closed loop transfer function is derived by analyzing the block diagram using the formula

$$\frac{\vartheta(s)}{D(s)} = \frac{\text{Forward Path}}{1 - \text{Loop Gain}}$$

$$\begin{aligned}
 G_c^{PD}(s) &= \frac{\vartheta(s)}{D(s)} = \frac{G_{\vartheta,D}(s)}{1 + G_{\vartheta,I} \cdot R(s)} \\
 &= \frac{\frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{[(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]}}{1 + \frac{I_1 \cdot \alpha_m}{[(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2]} \cdot (K_P + s \cdot K_D)} \\
 &= \frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{s^2 \cdot [I_2 \cdot (M + I_0) - I_1^2] + s \cdot (I_1 \cdot \alpha_m \cdot K_D) + I_1 \cdot \alpha_m \cdot K_P - I_1 \cdot g \cdot (M + I_0)} \\
 &= \frac{\frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{I_2 \cdot (M + I_0) - I_1^2}}{s^2 + s \cdot \frac{(I_1 \cdot \alpha_m \cdot K_D)}{I_2 \cdot (M + I_0) - I_1^2} + \frac{I_1 \cdot \alpha_m \cdot K_P - I_1 \cdot g \cdot (M + I_0)}{I_2 \cdot (M + I_0) - I_1^2}}
 \end{aligned} \tag{34}$$

5.3. Stability conditions (PD controller)

Asymptotic stability of a system is guaranteed if and only if all poles of the transfer function have strictly negative real parts. For a second-order characteristic polynomial, the Routh-Hurwitz stability criterion reduces to the necessary and sufficient condition that all coefficients of the polynomial must be nonzero and share the same sign. Therefore, considering the denominator of (34) the condition that K_p and K_d must satisfy is

$$\underbrace{[(M + I_0) \cdot I_2 - I_1^2] \cdot s^2 + (K_d \cdot I_1 \cdot \alpha_m) \cdot s + K_p \cdot I_1 \cdot \alpha_m - (M + I_0) \cdot I_1 \cdot g}_{>0}$$

Assuming a constant density, the first coefficient is greater than zero; thus, all the coefficients must be of positive sign:

$$\begin{aligned} K_d \cdot I_1 \cdot \alpha_m > 0 &\Rightarrow K_d > 0 \\ K_p \cdot I_1 \cdot \alpha_m - (M + I_0) \cdot I_1 \cdot g > 0 &\Rightarrow K_p > \frac{(M + I_0) \cdot g}{\alpha_m} \end{aligned}$$

5.4. PD control design

To obtain a more physically meaningful interpretation of the control parameters, it is convenient to express the coefficients K_p and K_d in terms of the natural frequency ω_c and the damping ratio ξ_c . To do so, the denominator must be similar to the denominator in the Nyquist form: $s^2 + 2 \cdot \xi_c \cdot \omega_c \cdot s + \omega_c^2$.

$$\begin{aligned} s^2 + \underbrace{\frac{(K_d \cdot I_1 \cdot \alpha_m)}{[(M + I_0) \cdot I_2 - I_1^2]} \cdot s}_{2 \cdot \xi_c \cdot \omega_c} + \underbrace{\frac{K_p \cdot I_1 \cdot \alpha_m - (M + I_0) \cdot I_1 \cdot g}{[(M + I_0) \cdot I_2 - I_1^2]}}_{\omega_c^2} & \\ 2 \cdot \xi_c \cdot \omega_c = \frac{(K_d \cdot I_1 \cdot \alpha_m)}{[(M + I_0) \cdot I_2 - I_1^2]} \rightarrow K_d = \frac{2 \cdot \xi_c \cdot \omega_c \cdot [(M + I_0) \cdot I_2 - I_1^2]}{I_1 \cdot \alpha_m} > 0 & (35) \\ \omega_c^2 = \frac{K_p \cdot I_1 \cdot \alpha_m - (M + I_0) \cdot I_1 \cdot g}{[(M + I_0) \cdot I_2 - I_1^2]} \rightarrow K_p = \frac{\omega_c^2 \cdot [(M + I_0) \cdot I_2 - I_1^2] + (M + I_0) \cdot I_1 \cdot g}{I_1 \cdot \alpha_m} > \frac{(M + I_0) \cdot g}{\alpha_m} & \\ \Rightarrow \begin{cases} \omega_c^2 > 0 \\ \xi_c \cdot \omega_c > 0 \end{cases} \rightarrow \begin{cases} \omega_c > 0 \\ \xi_c > 0 \end{cases} & \end{aligned}$$

The synthesis of the PD controller was carried out analytically. During the design phase, the denominator of the closed-loop transfer function was modeled as a pair of complex conjugate poles, expressed in terms of ξ_c and ω_c . By equating the coefficients of this target denominator with those derived from the system's analytical model, exact values for K_p and K_d were obtained, thus avoiding any form of approximation.

Consequently, the core of the design process consisted solely in selecting the appropriate values for ξ_c and ω_c . This choice was driven by specific desired dynamic requirements. In particular, the target criteria were a rise time $t_r < 0.15$ s and a settling time $t_s < 0.4$ s. Meeting these strict constraints ensures that the controller intervenes before the pendulum's inclination exceeds the validity bounds of the linearized model.

To satisfy these conditions, the parameters were set to $\xi_c = 0.7$ and $\omega_c = 15$ rad/s. Using these values alongside the given system parameters, it was possible to determine the following values for K_p and K_d :

$$\begin{cases} K_p = 27.2406 \\ K_d = 2.2493 \end{cases}$$

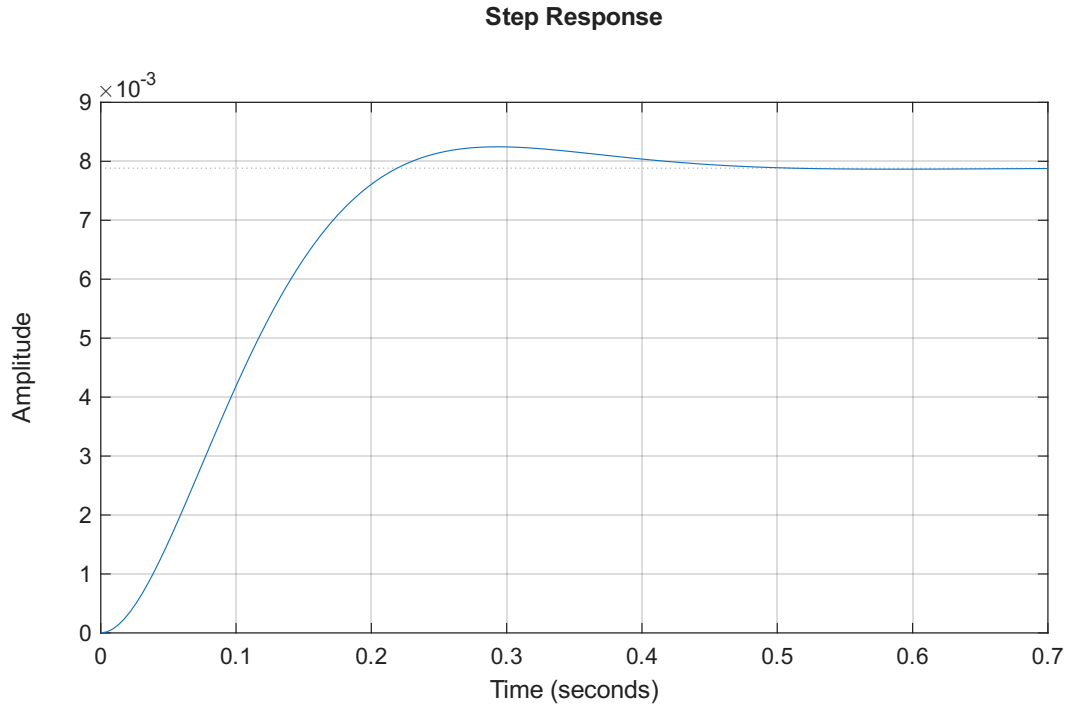


Figure 21: Step Response

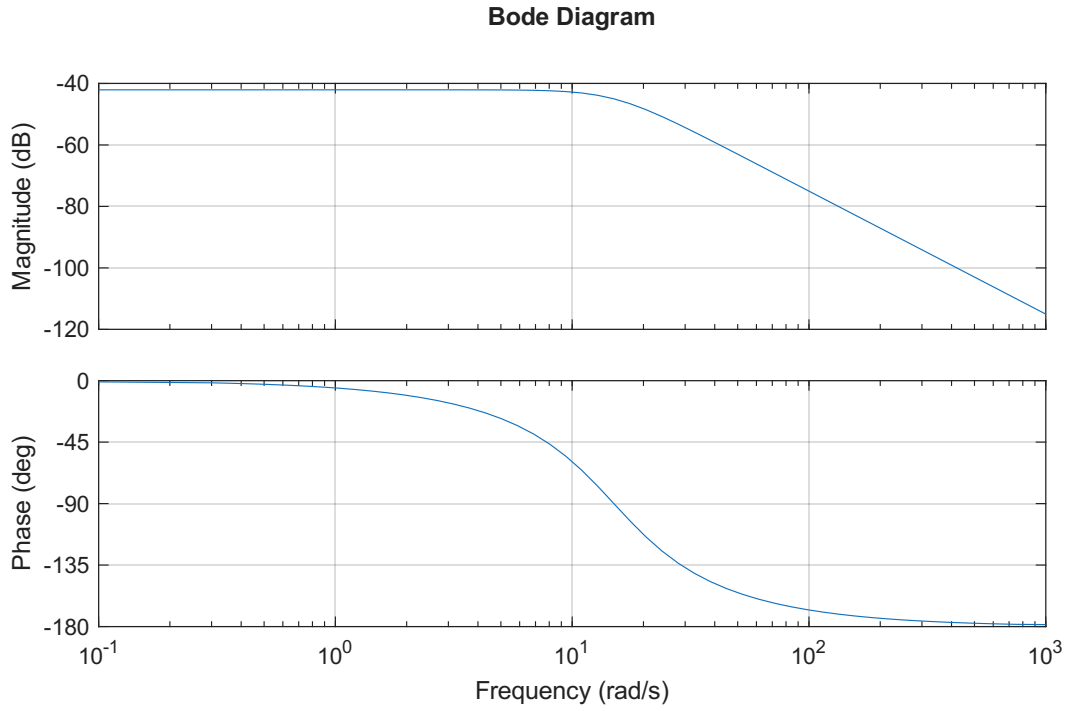


Figure 22: Bode Diagram

An analysis of the plots perfectly confirms the outcomes of the previous design phase. From the step response graph, it can be observed that both the rise time (t_r) and the settling time (t_s) are very small, allowing the system to react swiftly. Additionally, the chosen value of ξ_c ensures a very low overshoot, resulting in a well-damped system that rapidly converges to its steady-state value. Furthermore, evaluating the steady-state behavior anticipates the phenomenon that will be thoroughly analyzed in Task 5.5. Indeed, it is noticeable that the steady-state value upon which the system settles is strictly non-zero, foreshadowing the PD controller's structural inability to eliminate such steady-state error.

The Bode diagrams also validate the intended design. They show that at the frequency ω_c , the magnitude plot changes its slope, highlighting an effective capability to filter measurement noise. The role of the two complex conjugate poles is further corroborated by the phase plot, which exhibits a drop of 180° exactly around ω_c .

5.5. Steady-state error under constant disturbance

Using the final value theorem and assuming an asymptotically stable closed-loop system, the steady-state value of $\theta(t)$ is:

$$\begin{aligned} \lim_{s \rightarrow 0} s \cdot G_s^{PD} \cdot d(s) &= \lim_{s \rightarrow 0} \frac{[(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0] \cdot \bar{d}}{s^2 \cdot [I_2 \cdot (M + I_0) - I_1^2] + (I_1 \cdot \alpha_m \cdot K_d) \cdot s + I_1 \cdot \alpha_m \cdot K_p - I_1 \cdot g \cdot (M + I_0)} \\ &= \frac{[(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0] \cdot \bar{d}}{I_1 \cdot \alpha_m \cdot K_p - I_1 \cdot g \cdot (M + I_0)} \end{aligned}$$

By evaluating the limit, it can be observed that the steady-state error for a PD controller is strictly non-zero. However, analyzing the analytical expression of the steady-state error reveals how it can be minimized by tuning the parameters of the PD regulator.

Examining the numerator, it is evident that it depends primarily on the geometric parameters of the physical system, meaning it remains unaffected by variations in the controller. Conversely, the denominator is dependent on the proportional gain K_p . It can be noted that the denominator grows as K_p increases. Consequently, two main conclusions can be drawn:

- The irrelevance of the derivative term: K_d does not appear in the steady-state error expression, confirming it has no impact on the system's static accuracy.
- The role of the proportional gain: recalling the analytical expression for K_p , it can be seen that K_p grows as the natural frequency ω_c increases. Therefore, increasing the natural frequency reduces the steady-state error, but simultaneously makes the system more sensitive to high-frequency noise.

$$K_p = \frac{\omega_c^2 \cdot [(M + I_0) \cdot I_2 - I_1^2] + (M + I_0) \cdot I_1 \cdot g}{I_1 \cdot \alpha_m}$$

5.6. Closed-loop time-domain response

The following plots illustrate the time histories of the cart position, the pendulum angle, and the motor current when the frictionless system is subjected to a constant disturbance between 1 s and 3 s, under zero initial conditions.

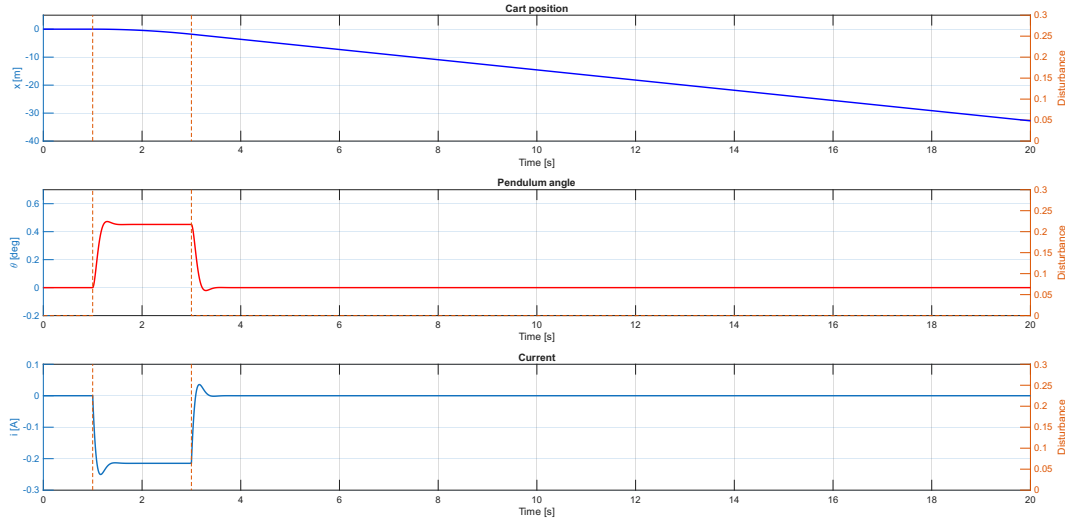


Figure 23: Closed-loop time-domain response (PD)

The plots perfectly illustrate the characteristics of the described system: a linearized frictionless model where a PD control has been implemented on θ .

The graphs highlight the regulator's performance, as it successfully stabilizes θ both during the disturbance interval and in its aftermath. The speed of this convergence serves as an indicator of the efficiency of the designed PD controller. This behavior is closely coupled with the control current, which varies during the stabilization of the rod's inclination and settles at zero once the rod is restored to the upright position.

The most distinctive feature, however, is the cart position, which fully captures the defining characteristic of this task: the absence of friction. During the disturbance, the cart shifts to the left to balance the rod's inclination, responding to the control action. After the 3-second mark, the current ceases to drive the movement; however, due to the lack of friction and the conservation of momentum, the cart is unable to decelerate or stop. Consequently, it continues to move in the negative x -direction at a constant velocity due to the lack of a control loop for the cart's position.

5.7. Effect of friction (model mismatch)

This section highlights the limitations of the PD controller in the presence of viscous friction. Specifically, while the controller and system parameters remain identical to those in the previous task, viscous friction has now been introduced into the model, creating a significant model mismatch.

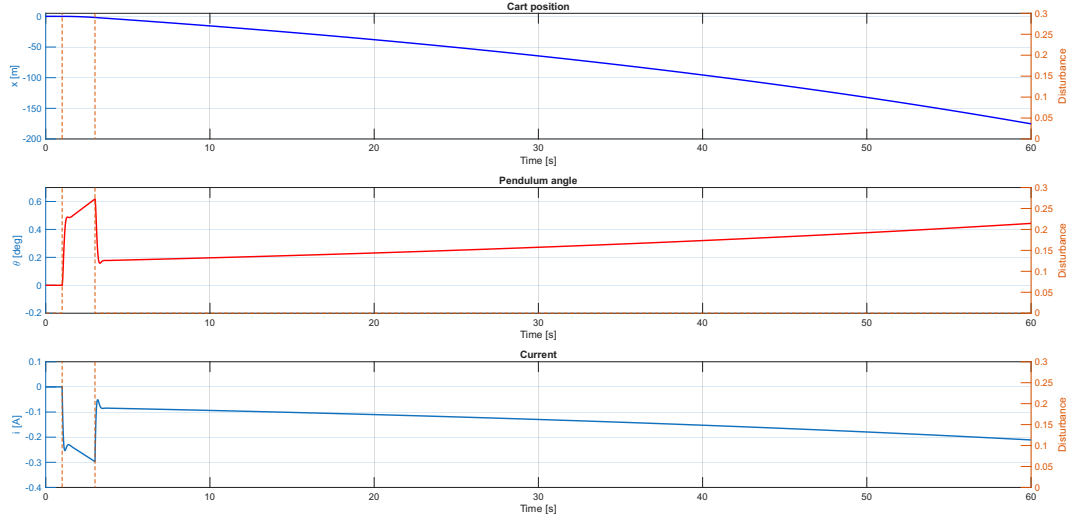


Figure 24: Effect of viscous friction on PD controller

The presented plots clearly illustrate the system's behavior in the presence of friction. Regarding the pendulum angle, friction prevents its stabilization during the interval when the disturbance is active. For the same reason, once the disturbance ceases, the angle θ begins to increase unboundedly over time. This directly impacts both the motor current and the cart position. Indeed, it can be observed that to compensate for the slow fall of the rod, the cart is forced to accelerate, moving increasingly faster. Consequently, the current must also grow progressively more negative over time to provide the necessary acceleration for the cart.

Ultimately, these graphs demonstrate that the PD controller is unable to compensate for the effects of friction, thus failing to regulate the rod's inclination. This failure propagates to the current and the cart position, which both grow over time. Eventually, this regulator will reach a condition where the linear model ceases to be valid and the system diverges, as evidenced by the plots.

An analysis of the impact of friction on the controller's performance reveals that the contribution of viscous friction yields fundamentally different effects depending on the system's dynamics. Two distinct scenarios can be evaluated:

- Scenario 1: Controller operating around a stable equilibrium point. Under these conditions, the effect of friction is beneficial for the control action, as it naturally damps the oscillations around the equilibrium position, thereby reducing the overall control effort required. For instance, in a standard pendulum subjected to a disturbance, viscous friction attenuates the oscillations, restoring the equilibrium condition more rapidly.
- Scenario 2: Controller operating around an unstable equilibrium point This is the specific scenario characterizing the system under analysis, as evidenced by the provided plots. Here, friction counteracts the regulatory action, imposing a significantly greater control effort to restore the pendulum to the vertical position. Since the target equilibrium is inherently unstable, the pendulum naturally tends to diverge from it; consequently, friction opposes the rapid corrective movements required by the cart, thus degrading the overall efficiency of the regulator.

5.8. Closed-loop model (PID controller)

The Proportional-Integrative-Derivative controller presents a form like: $\frac{K_i}{s} + K_p + K_d \cdot s$. In order to obtain the closed loop transfer function, the path is the same as what has been done in Task 5.2

$$\begin{aligned} G_s^{PID} &= \frac{\vartheta(s)}{D(s)} = \frac{G_{\vartheta,D}(s)}{1 + G_{\vartheta,I}(s) \cdot R(s)} \\ &= \frac{s \cdot [(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0]}{s \cdot [(M + I_0) \cdot (I_2 \cdot s^2 - I_1 \cdot g) - I_1^2 \cdot s^2] + I_1 \cdot \alpha_m \cdot [K_d \cdot s^2 + K_p \cdot s + K_i]} \\ &= \frac{s \cdot [(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0]}{s^3 \cdot [(M + I_0) \cdot I_2 - I_1^2] + s^2 \cdot I_1 \cdot \alpha_m \cdot K_d + s \cdot [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] + I_1 \cdot \alpha_m \cdot K_i} \end{aligned}$$

5.9. Steady-state error under constant disturbance (PID controller)

By applying the final value theorem, it can be demonstrated that the integral action of a PID controller drives the steady-state error asymptotically to zero:

$$\begin{aligned} \lim_{s \rightarrow 0} s \cdot G_c^{PID}(s) \cdot \frac{\bar{D}}{s} &= \lim_{s \rightarrow 0} \frac{s \cdot [(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0] \cdot \bar{D}}{s^3 \cdot [(M + I_0) \cdot I_2 - I_1^2] + s^2 \cdot I_1 \cdot \alpha_m \cdot K_d + s \cdot [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] + I_1 \cdot \alpha_m \cdot K_i} \\ &= \lim_{s \rightarrow 0} \frac{[(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0] \cdot \bar{D}}{s^2 \cdot [(M + I_0) \cdot I_2 - I_1^2] + s \cdot I_1 \cdot \alpha_m \cdot K_d + [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] + \underbrace{\frac{I_1 \cdot \alpha_m \cdot K_i}{s}}_{\rightarrow \infty}} \\ &\rightarrow 0 \end{aligned}$$

This result clearly highlights the fundamental difference between the previous PD regulator and the newly designed PID controller. The latter's ability to achieve zero steady-state error is reflected in its behavior during the response to a constant disturbance.

5.10. Stability conditions (PID controller)

Regarding the stability of the system, the considerations detailed in Section 5.3 remain applicable. However, as the characteristic polynomial is of the third order, the Routh-Hurwitz criterion regarding the sign of the coefficients provides only a necessary, but not sufficient, condition for stability. Consequently, the Routh table must be constructed to evaluate the stability conditions explicitly.

$$s^3 \cdot [(M + I_0) \cdot I_2 - I_1^2] + s^2 \cdot I_1 \cdot \alpha_m \cdot K_d + s \cdot [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] + I_1 \cdot \alpha_m \cdot K_i$$

Routh table

$[(M + I_0) \cdot I_2 - I_1^2]$	$[-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p]$
$I_1 \cdot \alpha_m \cdot K_d$	$I_1 \cdot \alpha_m \cdot K_i$
$\star$	0
Δ	0

where

$$\star = - \frac{[(M + I_0) \cdot I_2 - I_1^2] \cdot I_1 \cdot \alpha_m \cdot K_i - [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] \cdot I_1 \cdot \alpha_m \cdot K_d}{I_1 \cdot \alpha_m \cdot K_d}$$

$$\Delta = I_1 \cdot \alpha_m \cdot K_i$$

Necessary and sufficient condition of asymptotic stability: first column coefficients not null and concordant in sign:

$$\begin{aligned}
& [(M + I_0) \cdot I_2 - I_1^2] > 0 \\
& I_1 \cdot \alpha_m \cdot K_d > 0 \\
& - \frac{[(M + I_0) \cdot I_2 - I_1^2] \cdot I_1 \cdot \alpha_m \cdot K_i - [-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p] \cdot I_1 \cdot \alpha_m \cdot K_d}{I_1 \cdot \alpha_m \cdot K_d} > 0 \\
& I_1 \cdot \alpha_m \cdot K_i > 0
\end{aligned}$$

After the substitutions, three conditions are derived:

$$\begin{aligned}
& K_d > 0 \\
& K_p > \frac{\frac{[(M + I_0) \cdot I_2 - I_1^2] \cdot K_i}{K_d} + (M + I_0) \cdot I_1 \cdot g}{I_1 \cdot \alpha_m} \\
& 0 < K_i < \frac{[K_p \cdot I_1 \cdot \alpha_m - (M + I_0) \cdot I_1 \cdot g] \cdot K_d}{[(M + I_0) \cdot I_2 - I_1^2]}
\end{aligned}$$

5.11. PID control design

As demonstrated for the PD controller, it is advantageous to express the transfer function's denominator in terms of ω_c and ξ_c . Since the PID controller results in a third-order denominator, it is necessary to add a fast real pole to the pair of complex conjugate poles already established for the PD case. This third pole must be significantly faster than the dominant pair to ensure it does not affect the overall system dynamics. The denominator can therefore be expanded as follows:

$$(s^2 + 2 \cdot \xi_c \cdot \omega_c \cdot s + \omega_c^2) \cdot (s + p_3) = s^3 + s^2 \cdot (2 \cdot \xi_c \cdot \omega_c + p_3) + s \cdot (\omega_c^2 + 2 \cdot \xi_c \cdot \omega_c \cdot p_3) + \omega_c^2 \cdot p_3$$

It is now possible to equate the coefficients of this polynomial with those of the transfer function denominator, proceeding in an analogous manner to the PD controller.

$$\begin{cases}
2 \cdot \xi_c \cdot \omega_c + p_3 = \frac{I_1 \cdot \alpha_m \cdot K_d}{(M + I_0) \cdot I_2 - I_1^2} \\
\omega_c^2 + 2 \cdot \xi_c \cdot \omega_c \cdot p_3 = \frac{-(M + I_0) \cdot I_1 \cdot g + I_1 \cdot \alpha_m \cdot K_p}{(M + I_0) \cdot I_2 - I_1^2} \\
\omega_c^2 \cdot p_3 = \frac{I_1 \cdot \alpha_m \cdot K_i}{(M + I_0) \cdot I_2 - I_1^2}
\end{cases}$$

Solving the system for the gains K_p , K_d , and K_i :

$$\begin{cases}
K_d = \frac{(2 \cdot \xi_c \cdot \omega_c + p_3) \cdot [(M + I_0) \cdot I_2 - I_1^2]}{I_1 \cdot \alpha_m} \\
K_p = \frac{(\omega_c^2 + 2 \cdot \xi_c \cdot \omega_c \cdot p_3) \cdot [(M + I_0) \cdot I_2 - I_1^2] + (M + I_0) \cdot I_1 \cdot g}{I_1 \cdot \alpha_m} \\
K_i = \frac{\omega_c^2 \cdot p_3 \cdot [(M + I_0) \cdot I_2 - I_1^2]}{I_1 \cdot \alpha_m}
\end{cases}$$

The values of the three parameters K_p , K_d , and K_i therefore depend on the values chosen for the three variables ξ_c , ω_c , and p_3 , and consequently on the location of the two complex conjugate poles and the real pole. Maintaining continuity with the previous study of the PD controller, it was decided to keep ξ_c and ω_c unchanged. This fixed locations of the two complex conjugate poles, which have a real part of approximately -10.5. As previously mentioned, the third pole must be significantly faster than the dominant pair to avoid altering the overall system performance, thereby maintaining a good response time and minimal overshoot. By placing a third real pole with more negative real part than that of the complex conjugate poles, the latter remain the dominant due to their proximity to the imaginary axis. For this reason, a value of $p_3 = 20$ was chosen.

However, the real pole p_3 cannot be placed arbitrarily far to the left in the complex plane. Since the integral gain K_i is directly proportional to the magnitude of p_3 , such a choice would generate an excessively high K_i . Analytically, an oversized integral gain introduces a severe phase lag that reduces the system's phase margin, nullifying the stabilizing effect of the derivative action. Furthermore, from a physical perspective, an exorbitant K_i would demand a disproportionate control effort even for minor errors, inevitably leading to actuator saturation potential instability.

Step Response

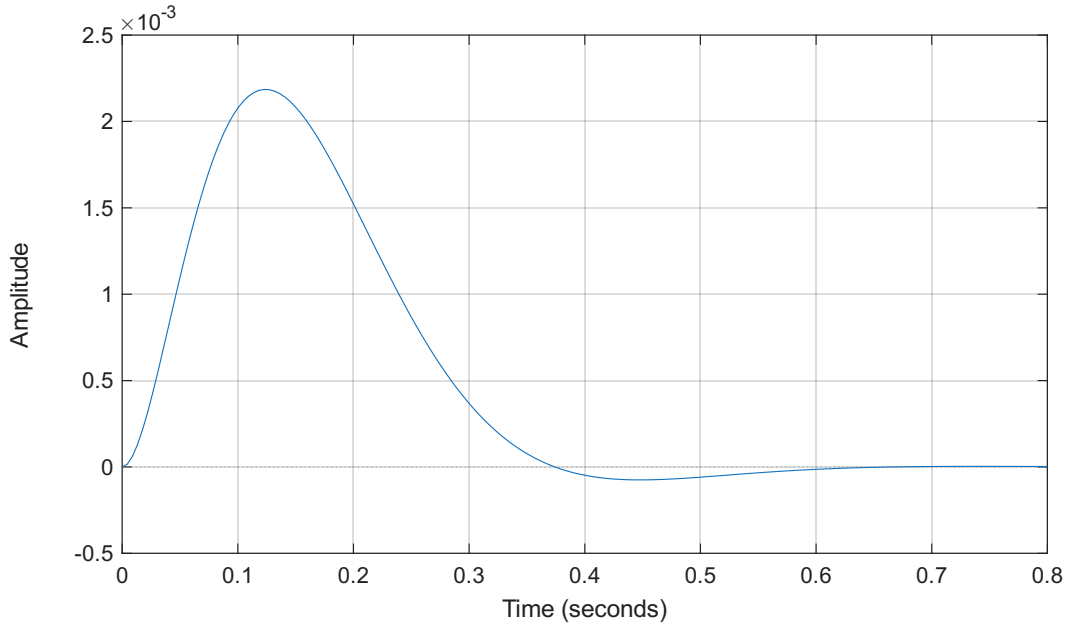


Figure 25: Step response

The step response plot clearly illustrates this behavior. Indeed, the system exhibits a minimal overshoot (approximately $2.2 \cdot 10^{-3}$) and settles to its steady-state value within 0.6 s. The most significant aspect is that the steady-state response value is exactly zero, confirming that the PID controller successfully eliminates steady-state error.

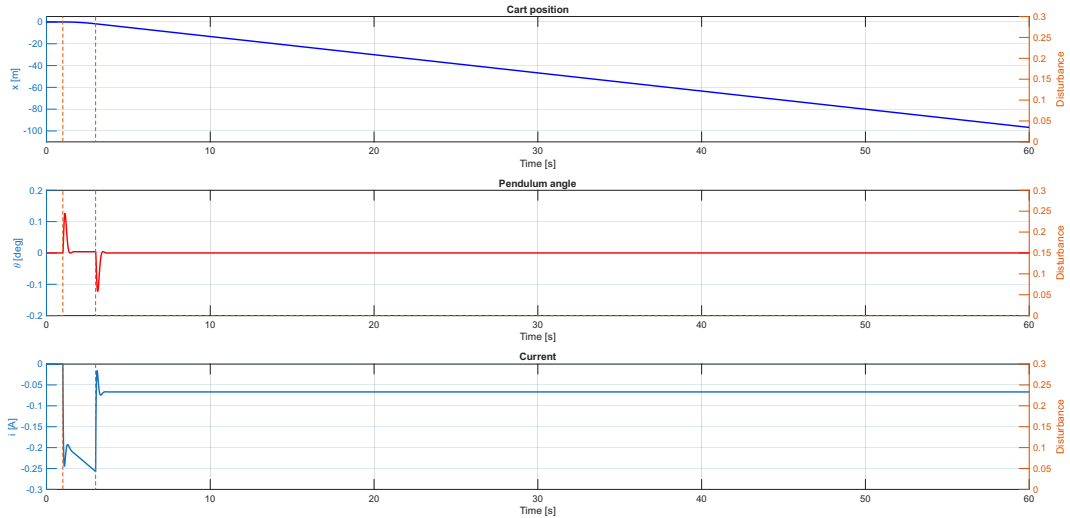


Figure 26: Closed-loop time-domain response (PID)

The plots clearly illustrate the full implications of employing a PID regulator for the control of θ . Regarding the angle, it is successfully stabilized both during and after the disturbance. The main difference compared to the PD controller is that the pendulum is forced back to the upright position even while the disturbance is active.

Another significant implication concerns the control current's behavior. Whereas with the PD controller the current returned to zero after regulating the rod's inclination for the second time, here it settles at a constant, slightly negative value. This is dictated by the intrinsic characteristics of the regulator: the integral action "integrates" the angular error history, accumulating a non-zero value to counteract the constant disturbance.

This peculiarity directly affects the cart's kinematics. Driven by a constant steady-state current, which translates to a constant force applied to the frictionless mass, the cart experiences a constant acceleration. Consequently, it can neither stop nor slow down, but will continue to drift indefinitely in the negative x -direction along a parabolic trajectory.

6. State Feedback Control

To determine the controllability and observability matrices, the linearized system matrices are provided below:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-c \cdot I_2}{\Delta} & \frac{I_1^2 \cdot g}{\Delta} & \frac{-b \cdot I_1}{\Delta} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{-c \cdot I_1}{\Delta} & \frac{(M + I_0) \cdot I_1 \cdot g}{\Delta} & \frac{-(M + I_0) \cdot b}{\Delta} \end{bmatrix} \quad B_i = \begin{bmatrix} 0 \\ \frac{I_2 \cdot \alpha_m}{\Delta} \\ 0 \\ \frac{I_1 \cdot \alpha_m}{\Delta} \end{bmatrix} \quad B_d = \begin{bmatrix} 0 \\ \frac{I_1 \cdot \alpha_1 - I_2 \cdot \alpha_0}{\Delta} \\ 0 \\ \frac{(M + I_0) \cdot \alpha_1 - I_1 \cdot \alpha_0}{\Delta} \end{bmatrix} \quad (36)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (37)$$

6.1. Controllability analysis

$$\mathcal{C} = \begin{bmatrix} B_i & AB_i & A^2B_i & A^3B_i \end{bmatrix} \quad (38)$$

By using the MATLAB command `ctrb(A, B_i)`, the controllability matrix can be computed. By verifying that the latter has full rank, it can be concluded that the system is fully controllable.

6.2. State feedback control law (pole placement)

The system is now considered to derive its closed-loop matrix. To achieve this, a state feedback control law of the form $i(t) = -K \cdot \vec{x}(t)$ is employed, where K is the state feedback gain vector.

$$\begin{cases} \vec{\dot{x}}(t) = A \cdot \vec{x}(t) + B_i \cdot i(t) + B_d \cdot d(t) \\ i(t) = -K \cdot \vec{x}(t) \end{cases}$$

$$\vec{\dot{x}}(t) = A \cdot \vec{x}(t) + B_i \cdot (-K \cdot \vec{x}(t)) + B_d \cdot d(t) = (A - B_i \cdot K) \cdot \vec{x} + B_d \cdot d(t) \quad (39)$$

It is therefore possible to define a new closed-loop matrix, A_{cl} , for the system as:

$$A_{cl} = A - B_i \cdot K \quad (40)$$

Since the system is completely controllable, its dynamic behavior and stability can be modified using the pole placement technique. The design procedure consists of first selecting the desired closed-loop poles in the left-half of the complex plane to guarantee asymptotic stability and a fast transient response. Subsequently, the state feedback gain matrix K is computed to assign these target values as the eigenvalues of the closed-loop matrix A_{cl} .

In designing the controller via pole placement, four poles must be assigned. In this context, the most effective approach is generally to use one pair of complex conjugate poles alongside two strictly real poles (all with negative real parts). This combination ensures a fast transient response while keeping overshoots and oscillations strictly bounded.

An alternative approach could be assigning two pairs of complex conjugate poles to speed up the initial rise time. However, this is detrimental: the two pairs, oscillating at different frequencies, would superimpose. This leads to interferences between modes, characterized by higher peak values, a longer settling time, and highly unpredictable system dynamics. Furthermore, this would translate into an erratic physical movement of the cart and a significantly higher control effort.

Finally, one could opt for four strictly real poles to completely eliminate any oscillation. Although overshoots are avoided, this choice yields a sluggish and heavily damped initial response. In practice, this is highly critical for an inverted pendulum: a delayed reaction from the controller would allow the pendulum to fall too far from its upright equilibrium point, pushing the system outside the validity region of the linearized model.

Therefore, the most effective strategy consists of utilizing one complex conjugate pair alongside two strictly real poles. For the present problem, the desired poles have been assigned as follows:

$$\begin{cases} P1_c = -\xi_c \cdot \omega_c + j \cdot \omega_c \cdot \sqrt{1 - \xi_c^2} = -10.5 + j \cdot 10.71 \\ P2_c = -\xi_c \cdot \omega_c - j \cdot \omega_c \cdot \sqrt{1 - \xi_c^2} = -10.5 - j \cdot 10.71 \\ P3_c = -7 \\ P4_c = -9 \end{cases} \quad \text{using: } \begin{cases} \xi_c = 0.7 \\ \omega_c = 15 \text{ rad/s} \end{cases}$$

The system response is now analyzed under two distinct scenarios.

Case 1: System response to initial conditions in the absence of disturbances.

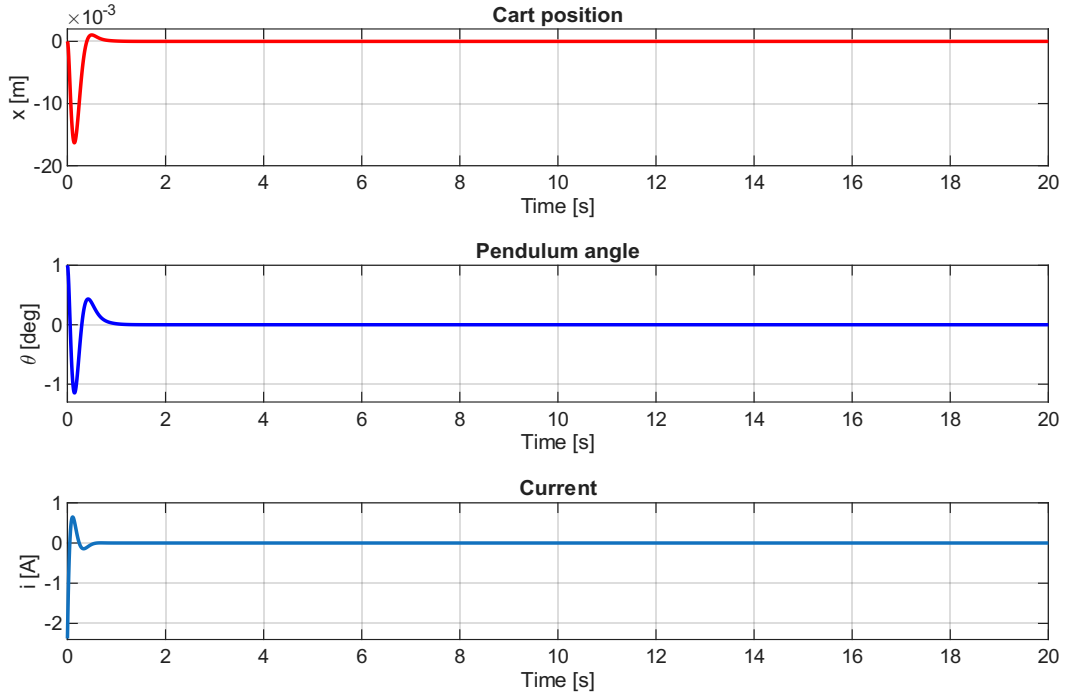


Figure 27: System response to initial conditions

In the unforced response to initial conditions, the desired characteristics sought through the pole placement design can be clearly observed. The system reacts rapidly with minimal oscillations. By analyzing the closed-loop matrix A_{cl} , it is possible to demonstrate the mathematical basis behind such a rapid settling time.

$$\vec{\dot{x}}(t) = A_{cl} \cdot \vec{x}(t)$$

The solution to the system under the specified initial conditions is given by:

$$\vec{x}(t) = e^{A_{cl} \cdot t} \cdot \vec{x}_0$$

The strictly negative eigenvalues of A_{cl} explain the fast decay of the system response towards the origin.

Case 2: System response to disturbances with zero initial conditions.

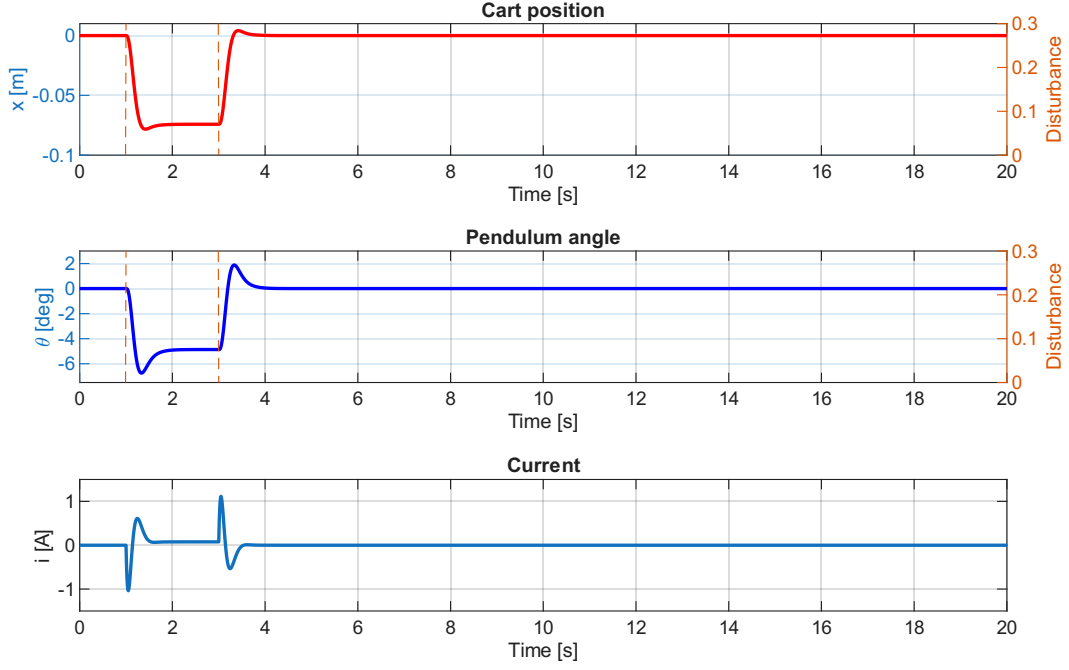


Figure 28: System response to disturbances

When the system is subjected to a disturbance under zero initial conditions, the behavior is significantly different. During the disturbance interval, the system stabilizes around a new equilibrium point for both the rod and the cart. This demonstrates the high efficiency and rapid response of the designed state-feedback system, which manages to reach a steady state even before the disturbance ends. Once the disturbance concludes, the system settles back to its original upright equilibrium position.

It is worth delving deeper into an important aspect of this behavior. As previously mentioned, during the disturbance interval, the system reaches an alternative equilibrium position. In this state, the rod maintains a constant tilt; the resulting shift in its center of mass effectively balances the horizontal force exerted by the disturbance. Consequently, the angle does not return to zero but remains constant until the disturbance is removed.

This behavior is entirely expected, as the control law implemented in the design, $i(t) = -K \cdot \vec{x}(t)$, is purely proportional to the state. Due to the absence of an integral component in the state-feedback architecture, the controller is unable to achieve a zero steady-state error when faced with a constant external disturbance. This result confirms that while state feedback is excellent for pole placement and stabilization, it lacks the inherent disturbance rejection capabilities of an integral action.

6.3. State observer design

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix} \quad (41)$$

Analogous to the controllability analysis, the full observability of the system is verified in MATLAB by constructing the observability matrix using the `obsv(A,C)` function and confirming that it has full row rank.

The design of the observer requires a careful selection of its poles (P_o) to ensure that the estimated states converge to the true states rapidly. According to control theory guidelines, the observer dynamics must be sufficiently faster than the controller dynamics. Therefore, the observer poles were placed approximately 2 to 6 times further to the left in the complex plane compared to the controller poles (P_c). This ensures that the estimation error decays to zero quickly enough to provide the controller with accurate state information before significant control action is exerted.

However, placing the observer poles arbitrarily far to the left was intentionally avoided. While extremely fast poles yield theoretically rapid convergence, they significantly increase the observer's bandwidth. In a physical scenario, a high-bandwidth

observer amplifies high-frequency sensor noise. This would result in an erratic control signal $i(t)$ that could easily lead to actuator saturation or mechanical wear. The chosen poles represent the optimal compromise between a fast transient response and robustness against measurement noise.

6.4. Dynamic compensator

Given that the system is fully observable, it is possible to design a state observer tasked with estimating the state variables from the available outputs. The following derivation presents the calculations required to analyze the system under the joint action of controller K and observer L .

$$\begin{cases} \vec{\tilde{x}}(t) = A \cdot \vec{x}(t) + B_i \cdot i(t) + B_d \cdot d(t) \\ \dot{\vec{\tilde{x}}}(t) = \hat{A} \cdot \vec{\tilde{x}}(t) + \hat{B}_i \cdot i(t) + L \cdot (\vec{y}(t) - \vec{\tilde{y}}(t)) \\ \vec{y}(t) = C \cdot \vec{x}(t) \\ \vec{\tilde{y}}(t) = C \cdot \vec{\tilde{x}}(t) \\ i(t) = -K \cdot \vec{\tilde{x}}(t) \end{cases} \quad (42)$$

$$\begin{cases} \vec{\tilde{x}}(t) = A \cdot \vec{x}(t) + B_i \cdot (-K \cdot \vec{\tilde{x}}) + B_d \cdot d(t) \\ \dot{\vec{\tilde{x}}}(t) = \hat{A} \cdot \vec{\tilde{x}}(t) + \hat{B}_i \cdot [-K \cdot \vec{\tilde{x}}(t)] + L \cdot (C \cdot \vec{x}(t) - C \cdot \vec{\tilde{x}}(t)) \end{cases}$$

$$\begin{bmatrix} \dot{\vec{\tilde{x}}} \\ \vec{\tilde{x}} \end{bmatrix} = \begin{bmatrix} A & -B_i \cdot K \\ L \cdot C & A - B_i \cdot K - L \cdot C \end{bmatrix} \cdot \begin{bmatrix} \vec{\tilde{x}}(t) \\ \vec{\tilde{x}}(t) \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} \cdot d(t) \quad (43)$$

It can be observed that in the resulting equations, only $\vec{\tilde{x}}(t)$ is directly affected by the presence of the disturbance. In fact, the observer does not perceive the disturbance a priori, sensing its effects only when a discrepancy in the outputs is detected. This explains the transient estimation error between the physical plant and the state reconstructed via the observer.

6.5. Closed-loop simulation (MATLAB)

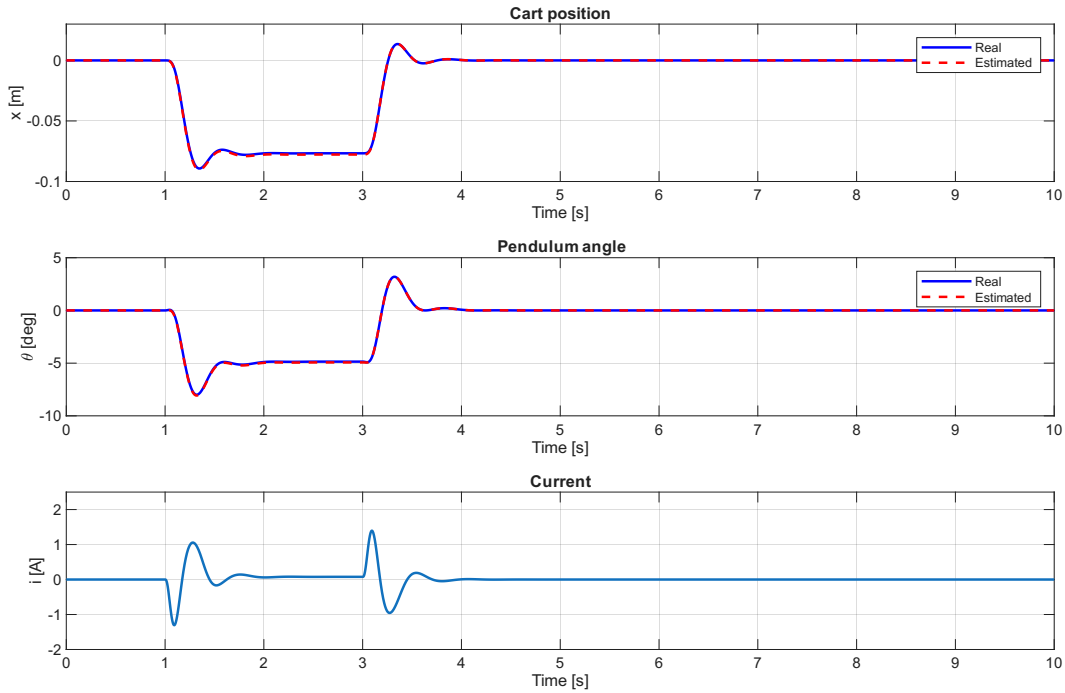


Figure 29: Comparison between the real and estimated systems

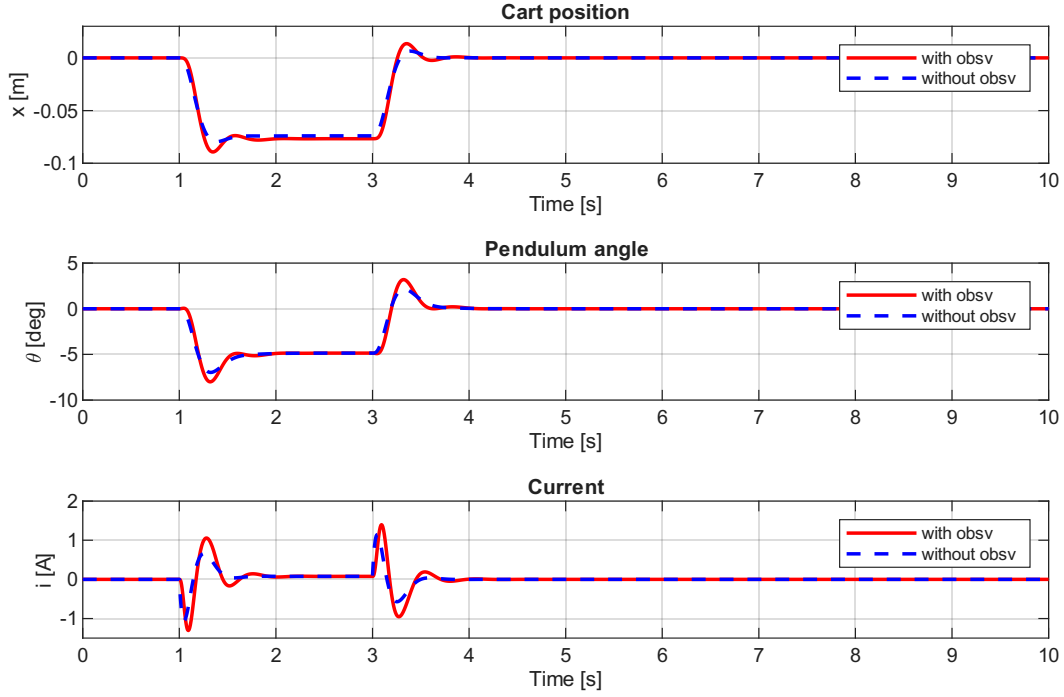


Figure 30: Comparison system with and without observer

As previously mentioned, the most significant difference between the physical plant and the reconstructed system lies in how the disturbance is processed. Specifically, the real system is directly subjected to the external disturbance, whereas the observer detects it indirectly and with a slight lag, inferring its presence solely from variations in the measured outputs.

This behavior is corroborated by the estimation error plots in Figure 31, which exhibit distinct peaks at the exact moments the disturbance is applied. This occurs because, while the physical plant is immediately affected by the disturbance, the observer-based compensator requires a transient phase to converge to the perturbed state. Consequently, in the very first instants following the external disturbance, the control effort $i(t)$ is calculated based on a temporarily inaccurate state estimation.

As observed in the time histories, this phenomenon results in a minor degradation of the transient response, specifically causing a slightly larger overshoot and a fractionally longer settling time compared to ideal state feedback. Furthermore, as the observer detects the disturbance and the compensator sharply corrects the trajectory, it generates a temporary spike in the control current. In this simulation, the current reaches a maximum peak of approximately 1.4 A. This value is entirely compatible with the operational limits of the DC motors typically employed in such applications, confirming that the selected poles represent an optimal balance between responsiveness and noise robustness, as discussed in Task 6.3:

$$\begin{cases} P1_o &= -\xi_c \cdot \omega_o + j \cdot \omega_o \cdot \sqrt{1 - \xi_c^2} = -21 + j \cdot 21.42 \\ P2_o &= -\xi_c \cdot \omega_o - j \cdot \omega_o \cdot \sqrt{1 - \xi_c^2} = -21 - j \cdot 21.42 \\ P3 &= -30 \\ P4 &= -40 \end{cases} \quad \text{using: } \begin{cases} \xi_c = 0.7 \\ \omega_o = 30 \text{ rad/s} \end{cases}$$

Therefore, the control action is fully feasible, and the risk of actuator saturation is effectively avoided, confirming both the robustness and the practical viability of the designed compensator.

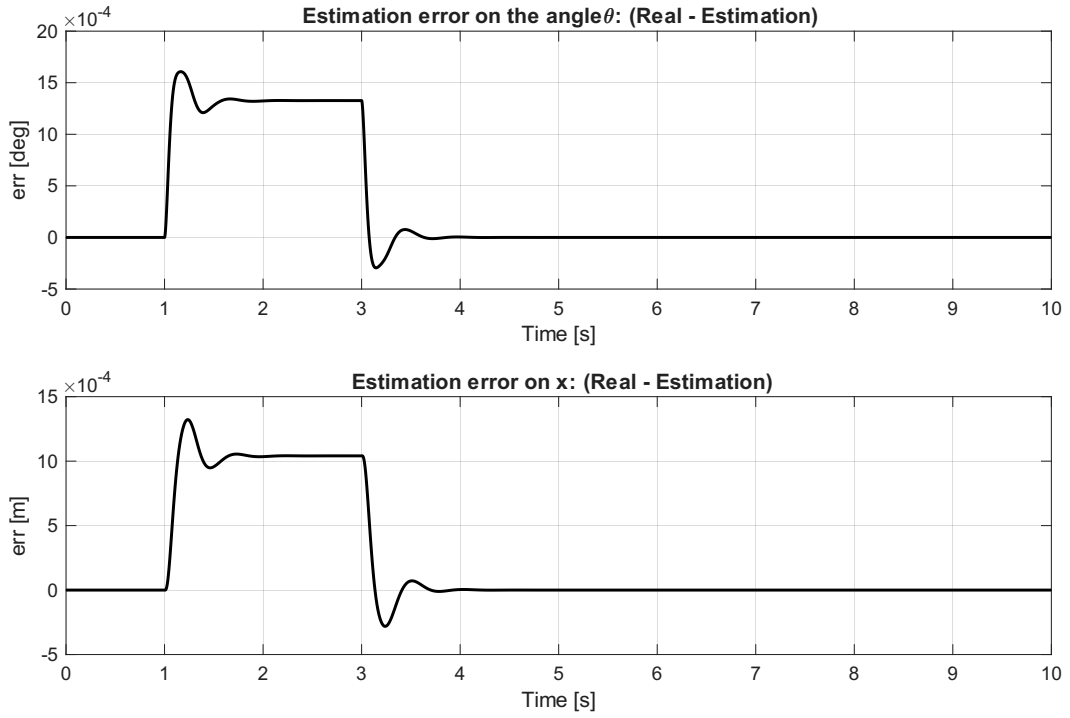


Figure 31: Estimation error between the real and estimated systems

6.6. Closed-loop simulation (Simulink)

Task 6.5 was replicated in Simulink via a MATLAB Function block. The resulting $x(t)$, $\theta(t)$, and $i(t)$ time histories match the MATLAB script output.

The comparison demonstrates a perfect overlap between the two responses, with identical dynamic behavior during the disturbance transient and the subsequent stabilization. This confirms the fundamental correctness of the closed-loop response for the system with the dynamic compensator derived in Task 6.4, under the same disturbance considered in Task 2.2, across both computational environments.

Any microscopic discrepancies are negligible and purely attributable to solver-specific numerical integration tolerances and dynamic time-stepping differences between the Simulink engine and the MATLAB `ode` solver. Similar considerations hold true for the comparison of the error graphs between the true and estimated states ($\vec{x}(t)$ vs $\hat{\vec{x}}(t)$) which exhibit identical profiles in both environments.

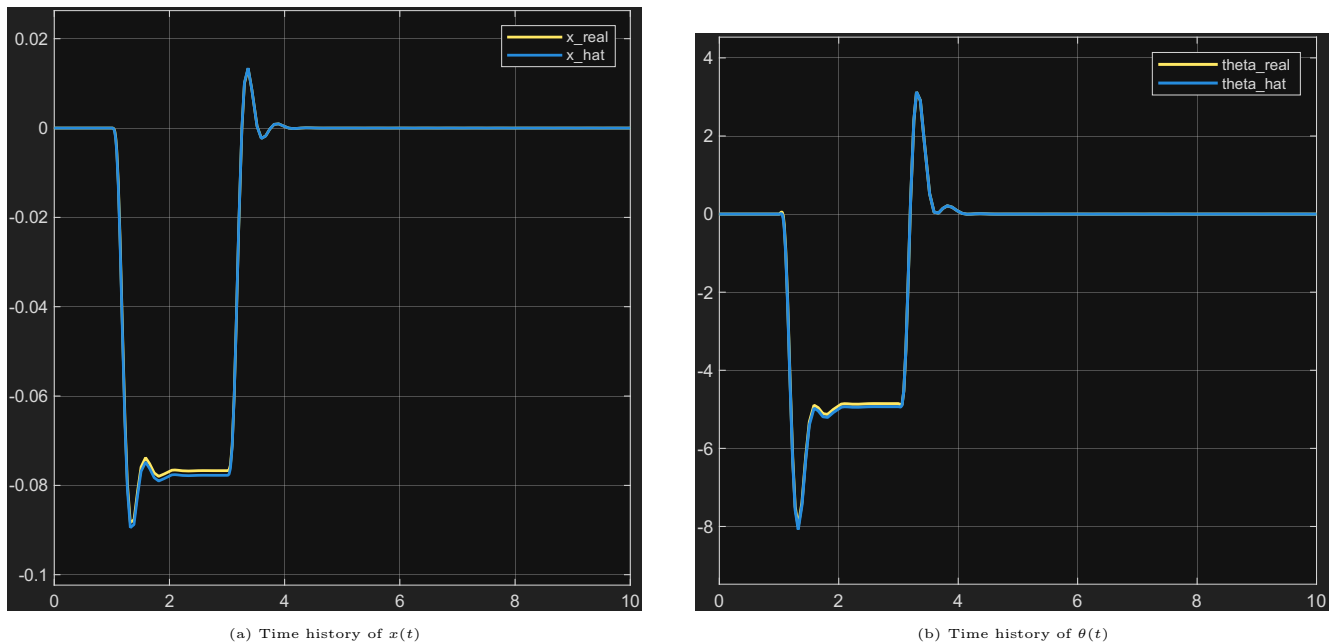


Figure 32: Simulated time responses of the system state variables.

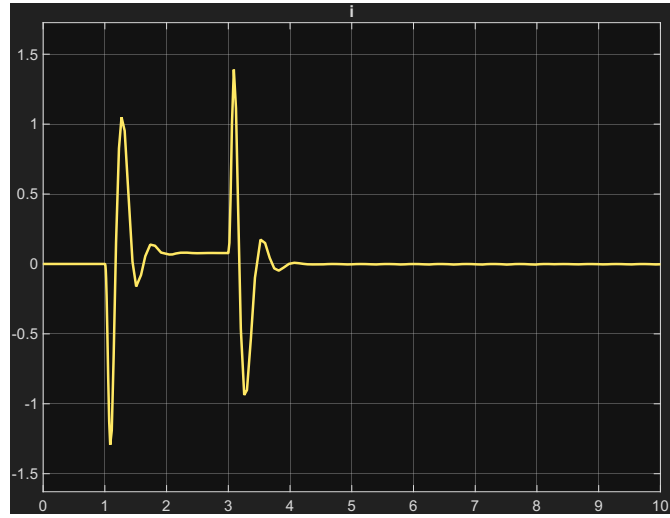
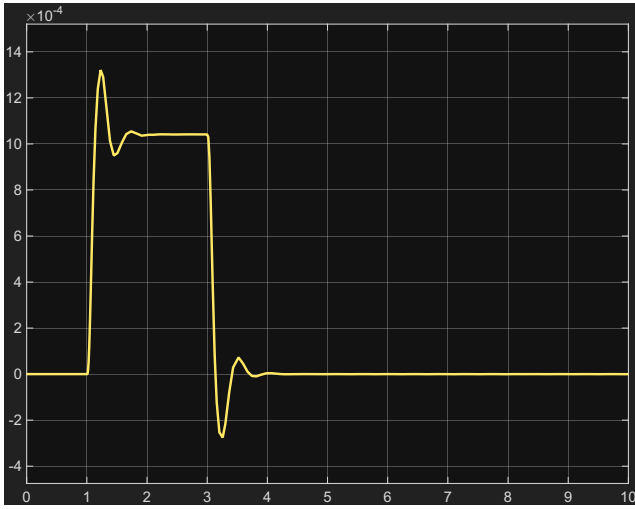
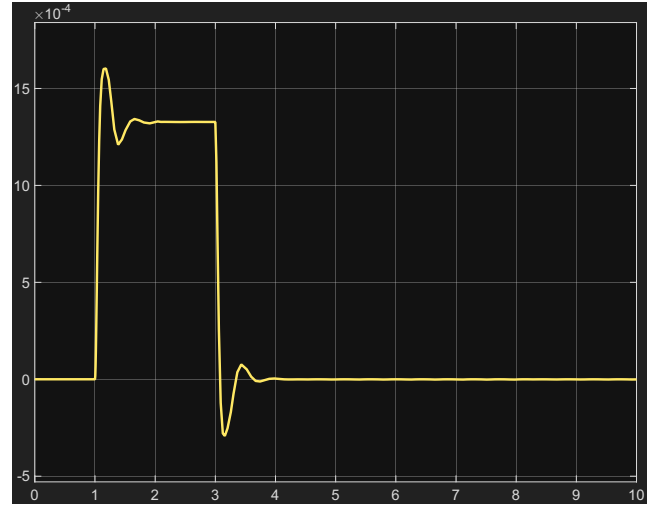


Figure 33: Time history of $i(t)$



(a) Estimation error $x(t) - \hat{x}(t)$



(b) Estimation error $\theta(t) - \hat{\theta}(t)$

Figure 34: Time history of the estimation errors for the cart position and pendulum angle.

7. Conclusions

This report addresses the modeling, analysis, and design of control architectures for an inverted pendulum on a cart. Following the assigned task-based structure, SISO controllers (PD and PID) were initially designed to stabilize only the pendulum angle, followed by the implementation of state-space techniques. Finally, the introduction of a state observer demonstrated how to effectively manage the structural limitations inherent in the system's kinematic drift.

Throughout the project, analytical and simulated results were continuously reviewed. Since software tools provide purely numerical outputs, it was essential to verify that every state and control variable remained physically consistent and within the operating tolerances of the real system. For this reason, the control effort, specifically the current drawn (control effort) by the motor, was closely monitored. Gain tuning and controller design were strictly limited by the hardware's physical saturation points. Consequently, controllers were rejected if they demanded unrealistic current peaks disproportionate to the system's mechanical inertia, even if they guaranteed theoretical asymptotic stability.

In conclusion, the proposed control architectures not only satisfy the dynamic requirements for transient response and disturbance rejection but also ensure practical feasibility by respecting the system's energy and physical constraints.